On Lifting AGM Security to AGM with Oblivious Sampling

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Abstract. Idealized models such as the Random Oracle Model and the Generic Group Model underpin much of modern provable security. The Algebraic Group Model (AGM) of Fuchsbauer, Kiltz, and Loss (CRYPTO 2018) attempts to bridge the gap to the standard model by forcing adversaries to justify every new group element via a linear representation in its inputs, and it was leveraged in many follow-up works. Lipmaa, Parisella, and Siim (TCC 2023) strengthened this framework to the AGM with Oblivious Sampling (AGMOS), which additionally lets attackers obtain random group elements without the knowledge of their discrete logarithms, reflecting the practical availability of hashing to the group via admissible encodings. Although several works already rely on AGMOS, the security arguments are still ad-hoc, and it is unclear whether an AGM proof automatically extends to this richer setting. We initiate a systematic study of this question. Our main result is a generic lifting theorem: For any assumption whose verification predicate satisfies a simple structural *lifting condition*, every AGM reduction remains sound in AGMOS under the Find Polynomial Representation (FPR) and Tensor Oracle Find Representation (TOFR) assumptions.

The lifting condition is met by a broad class of algebraic protocols. As illustrations, we lift from the AGM to the AGMOS (i) the knowledge soundness of KZG-like polynomial commitments and (ii) the security of the multivariate Adaptive Rational Strong Diffie–Hellman (ARSDH). Our results clarify the precise relationship between AGM and AGMOS and provide a black-box recipe for future work: once a proof is obtained in the AGM, one can often inherit AGMOS security with negligible additional effort. This both streamlines the use of AGMOS in forthcoming work and reinforces the confidence that schemes proven secure in the AGM remain robust when oblivious sampling is available.

Keywords: Algebraic Group Model · Oblivious Sampling · AGM · AGMOS · Lifting Security Proofs.

1 Introduction

Idealized models have become common in modern provable security. By idealizing the properties of some primitives or adversaries, we can argue rigorously about schemes that we currently cannot analyze in the standard model, providing at least some level of assurance about their security. Classic examples include the Random Oracle Model (ROM, [BR93]) and its quantum variant QROM [BDF+11] for hash functions, and Generic Group Model (GGM, [Sho97, Fis00, Mau05]) for hard problems over cyclic groups. Building on the GGM, Fuchsbauer, Kiltz, and Loss [FKL18] introduced the Algebraic Group Model (AGM). The AGM restricts the adversary's power by requiring that every new group element it outputs must come with an explicit algebraic representation in terms of the elements it has already seen. Intuitively, this restriction seems to capture a meaningful class of attackers that manipulate the input group elements in an "algebraic" manner. Moreover, it restores many extractor-based proof techniques and has enabled tight, natural security arguments for a wide array of schemes and protocols such as polynomial commitment schemes [CHM+20, BDFG21], SNARKs [MBKM19], or signature schemes [FPS20] (see also Zhang, Zhou, and Katz [ZZK22] for additional examples).

A technical limitation of the AGM is that it does not allow the adversary to sample group elements "obliviously," i.e., without knowing their discrete logarithms. However, it is well known that, in practice, the so-called admissible encodings enable such oblivious sampling without the need for any "non-algebraic" operations on group elements (see, e.g., [BF01, Ica09, WB19]). To mend this technical gap, Lipmaa, Parisella, and Siim [LPS23] extended the AGM and proposed the AGM with Oblivious Sampling (AGMOS), in which the attacker also enjoys oracle access to freshly sampled, unlinkable group elements. When introducing the AGMOS, Lipmaa et al. also presented a streamlined AGMOS proof technique, using two novel assumptions: Find Polynomial Representation (FPR) and Tensor Oracle Find Representation (TOFR). They then illustrated the technique on several examples. Unsurprisingly, the AGMOS proofs are often extensions of AGM proofs. To quote Lipmaa et al. [LPS23], *"[...] in the case of more complicated AGM proofs, adding the AGMOS part to it is "relatively" easy compared to the AGM proof itself."* Given the general template of proofs in Lipmaa et al., we ask the following natural question:

Can we generically lift proofs of security from AGM to AGMOS?

Besides clarifying the relationship between the two models, a generic way of lifting the security from AGM to AGMOS, under reasonable assumptions, would greatly simplify the use of AGMOS in future work.

1.1 Our Contributions

We take the first steps towards answering our motivating question and show that, for a certain class of assumptions[3], an AGM security proof can be lifted to the AGMOS under minimal assumptions. Specifically, we present the following:

1. We identify a structural *lifting condition* on the implicit verification polynomial used in proofs of security in the AGMOS, which characterizes a set of assumptions whose AGM security proofs are liftable to AGMOS. We show that, under the lifting condition and assuming the FPR and TOFR hardness assumptions, any AGM reduction goes through verbatim in AGMOS.
2. We illustrate our lifting theorem on examples such as the extractability game for the KZG polynomial commitment [KZG10] or the multi-variate Adaptive Rational Strong Diffie–Hellman assumption [BDH+25] used recently in the context of extractability of variants of KZG in the standard model.

1.2 Technical Overview

In this subsection, we provide a technical overview of our results. We begin by recalling the standard view of AGM security proofs through verification polynomials. We then explain how this viewpoint changes in the

[3] As [LPS23], by "assumption" we mean "an assumption or the security of a protocol."

AGMOS, and which additional cases arise. Finally, we state the single structural condition under which an AGM proof can be lifted to the AGMOS in a black-box way, and we illustrate it on a simple example.

We model each assumption as a game between a challenger and an AGMOS adversary. The challenger samples group elements and provides them to the adversary. The adversary then outputs group and field elements. Each output group element is accompanied by an explanation vector with respect to both the received inputs and the obliviously sampled elements. Finally, the challenger checks whether these outputs satisfy the winning condition. Throughout, we use bracket notation to denote group elements. Concretely, if $\mathbb{G}$ is an additive group and $G \in \mathbb{G}$ is its fixed generator, then for $a \in \mathbb{F}$ we write $[a] \coloneqq a \cdot G \in \mathbb{G}$. More generally, for a vector $\mathbb{a} \in \mathbb{F}^\ell$ we write $[\mathbb{a}] \in \mathbb{G}^\ell$ for the component-wise encoding, i.e., $[\mathbb{a}] \coloneqq ([a_1], \ldots, [a_\ell])$. We use the same notation for matrices, where $[\boldsymbol{A}]$ denotes component-wise encoding of $\boldsymbol{A} \in \mathbb{F}^{u \times v}$ into $\mathbb{G}^{u \times v}$.

AGM proofs via verification polynomials. In a typical AGM security proof of an assumption, one constructs a multivariate verification polynomial $V(\boldsymbol{X}) = V(X_1, \ldots, X_n)$. This polynomial is derived from the adversary's algebraic explanation vectors. Concretely, the adversary outputs a vector of group elements $[\mathbb{y}] = \boldsymbol{\gamma}[\mathbb{x}]$, where $\boldsymbol{\gamma}$ is a matrix whose rows are the explanation vectors, and $[\mathbb{x}]$ is the vector of input group elements provided by the challenger. The variables $X_1, \ldots, X_n$ correspond to the discrete logarithms of the group elements sampled by the challenger. The challenger derives the public inputs $[\mathbb{x}]$ from these discrete logarithms by applying known polynomial operations. As a result, whenever the adversary wins the game, it follows that $V(\boldsymbol{x}) = 0$, where $\boldsymbol{x}$ is the vector of discrete logarithms originally sampled by the challenger.

The proof then proceeds by distinguishing two cases.

Case A: When $V(\boldsymbol{X}) = 0$. One typically shows that this case is impossible.

Case X: When $V(\boldsymbol{X}) \neq 0$, but $V(\boldsymbol{x}) = 0$. One typically builds a reduction to a variant of the Power Discrete Logarithm (PDL) assumption.

Verification in the AGMOS. Recall that in the AGMOS setting, the adversary is more powerful than in the AGM. In addition to algebraic operations on its inputs, it has access to an oracle that returns randomly sampled group elements whose discrete logarithms are unknown to the adversary. Security proofs in the AGMOS still rely on verification polynomials. However, unlike in the AGM, the relevant polynomial may depend not only on the challenger's original group elements, but also on the obliviously sampled elements obtained from the oracle. Accordingly, the adversary outputs a vector of group elements $[\mathbb{y}]$ of the form

$$[\mathbb{y}] = \boldsymbol{\gamma}[\mathbb{x}] + \boldsymbol{\delta}[\mathbb{q}],$$

where $\boldsymbol{\gamma}$ explains the dependence on the challenger inputs $[\mathbb{x}]$, and $\boldsymbol{\delta}$ explains the dependence on the obliviously sampled elements $[\mathbb{q}]$.

The AGMOS verification polynomial and its decomposition. We denote the verification polynomial by $V(\boldsymbol{X}, \mathbb{Q})$. The vector of variables $\boldsymbol{X}$ corresponds to the discrete logarithms sampled by the challenger, from which the public inputs are derived by known polynomial operations. The vector of variables $\mathbb{Q}$ represents the (unknown) discrete logarithms of the obliviously sampled group elements. We further split the verification polynomial as

$$V(\boldsymbol{X}, \mathbb{Q}) = V^h(\boldsymbol{X}) + V^t(\boldsymbol{X}, \mathbb{Q}),$$

where $V^h(\boldsymbol{X})$ contains exactly the terms in $V(\boldsymbol{X}, \mathbb{Q})$ that involve only $\boldsymbol{X}$ and the constant term, and $V^t(\boldsymbol{X}, \mathbb{Q})$ is the remaining part that depends on $\mathbb{Q}$ (possibly together with $\boldsymbol{X}$).

Whenever the adversary succeeds in breaking the assumption, it must hold that $V(\boldsymbol{x}, \mathbb{q}) = 0$, where $\boldsymbol{x}$ are the challenger-sampled discrete logarithms and $\mathbb{q}$ are the (unknown) discrete logarithms of the obliviously sampled elements.

The four-case proof template in the AGMOS. A standard AGMOS proof distinguishes four cases.

Case A: When $V(\boldsymbol{X}, \mathbb{Q}) = 0$. One typically shows that this case is impossible.

Case $\mathsf{X.1}$**: When** $V(\boldsymbol{X}, \mathbb{Q}) \neq 0$**, but** $V^t(\boldsymbol{X}, \mathbb{Q}) = 0$**.** One is effectively in the AGM regime, since the $\mathbb{Q}$-dependent part vanishes identically.

Case $\mathsf{X.2}$**: When** $V^t(\boldsymbol{X}, \mathbb{Q}) \neq 0$**, but** $V^t(\boldsymbol{x}, \mathbb{Q}) = 0$**.** One typically builds a reduction to the FPR assumption (and hence to PDL).

Case Q**: When** $V^t(\boldsymbol{x}, \mathbb{Q}) \neq 0$**.** One builds a reduction to the TOFR assumption.

The auxiliary assumptions FPR *and* TOFR. The FPR assumption (see Definition 4) automates the PDL-style reasoning in cases $\mathsf{X.1}$ and $\mathsf{X.2}$. It states that it is hard to find a non-zero multivariate polynomial $f(\boldsymbol{X})$ that evaluates to zero at the given trapdoor input. For a reduction from FPR to PDL, see [LPS23]. The TOFR assumption, introduced in [LPS23], states that for an adversary with access to an obliviously sampling oracle, it is infeasible to find any non-trivial linear relation among the discrete logarithms of sampled group elements.

Relating AGM and AGMOS proofs. When comparing AGM and AGMOS proofs for the same security game, it is natural to compare their verification polynomials. The AGM verification polynomial $V(\boldsymbol{X})$ corresponds exactly to the $\boldsymbol{X}$-only component $V^h(\boldsymbol{X})$ of the AGMOS verification polynomial $V(\boldsymbol{X}, \mathbb{Q})$. Intuitively, $V^h(\boldsymbol{X})$ captures the part of the winning condition that is already present in the AGM, while $V^t(\boldsymbol{X}, \mathbb{Q})$ captures the new algebraic influence of obliviously sampled elements.

The lifting idea. Cases $\mathsf{X.2}$ and Q are handled by generic reductions to FPR and TOFR, respectively. Thus, to lift an AGM proof to the AGMOS, it remains to control cases A and $\mathsf{X.1}$. In these cases, we want to argue that the AGMOS adversary cannot gain any advantage from oblivious sampling, and therefore behaves like an AGM adversary. If this holds, then the original AGM proof directly handles the remaining cases, and the only additional assumptions needed are precisely those that cover cases $\mathsf{X.2}$ and Q.

A single structural condition. We formalize the requirement that oblivious sampling is useless in cases A and $\mathsf{X.1}$ as a condition on the game. Informally, if the $\mathbb{Q}$-dependent part $V^t(\boldsymbol{X}, \mathbb{Q})$ vanishes identically, then the adversary must not be able to use any obliviously sampled elements at all. Concretely, we require that

$$V^t(\boldsymbol{X}, \mathbb{Q}) = 0 \implies \boldsymbol{\delta} = \boldsymbol{0}.$$

This *lifting condition* ensures that whenever V^t vanishes, the adversary's explanation uses only challenger-provided inputs, exactly as in the AGM. Under this condition, together with FPR and TOFR (covering cases $\mathsf{X.2}$ and Q), we show that any AGM security proof can be lifted generically to the AGMOS. Importantly, the lifting condition is easy to verify in applications. After expanding $V^t(\boldsymbol{X}, \mathbb{Q})$, the statement $V^t(\boldsymbol{X}, \mathbb{Q}) = 0$ amounts to a simple system of linear or quadratic equations in the coefficients. Thus, checking whether non-zero solutions exist is typically straightforward.

Illustration on CDH. We illustrate the lifting approach on the CDH assumption. In the AGM, an algebraic CDH adversary $\mathcal{A}$, on input $[1, a, b]$, outputs $[ab]$ together with coefficients $\gamma_1, \gamma_2, \gamma_3$ such that

$$[ab] = \gamma_1[1] + \gamma_2[a] + \gamma_3[b].$$

A DLOG reduction can embed its challenge as $[a]$, sample b, run $\mathcal{A}$, and form the verification polynomial

$$V(X) = \gamma_1 + \gamma_2 X + \gamma_3 b - Xb.$$

If $\mathcal{A}$ succeeds then $V(a) = 0$. If V is not the zero polynomial, the reduction recovers a from V, which proves AGM security of CDH under DLOG.

In the AGMOS, the adversary may additionally use obliviously sampled elements, and the verification polynomial becomes

$$V(X, \mathbb{Q}) = \gamma_1 + \gamma_2 X + \gamma_3 b - Xb + \sum_{i=1}^{\mathsf{ql}_1} \delta_i \mathbb{Q}_i.$$

Here $V^t(\mathbb{Q}) = \sum_{i=1}^{\mathsf{ql}_1} \delta_i \mathbb{Q}_i$. Under the TOFR assumption, the event $V^t(\mathbb{Q}) \neq 0$ can only occur with negligible probability. Moreover, the lifting condition holds trivially, since $V^t(\mathbb{Q}) = 0$ implies $\delta_i = 0$ for all i, and hence $\boldsymbol{\delta} = \mathbf{0}$. Thus, assuming AGM security of CDH and TOFR, we conclude that CDH is also secure in the AGMOS. In this example, FPR is not needed, since the case X.2 does not arise.

1.3 Other Related Work

Although introduced only recently, the AGMOS has been recognized as an important refinement of the AGM. Some recent works used the AGMOS to analyze the security of their assumptions: Lipmaa, Parisella, and Siim [LPS24] analyzed their Adaptive Rational Strong Diffie-Hellman assumption in the AGMOS. Lipmaa [Lip24] proved the special soundness of the Polymath argument in AGMOS. Finally, Faonio, Fiore, and Russo [FFR24] claim that PLONK [GWC19] and Marlin [CHM+20] are knowledge-sound in the AGMOS, though their proofs are not detailed. Other recent works proved security in the AGM but explicitly identify the AGMOS as an important and natural model for future extension of their results: Dellepere et al. [DMS24] proved the knowledge soundness of their SNARKs Pari and Garuda in the AGM and stated that the results should extend to AGMOS, but they left a formal proof for future work. Campanelli et al. [CFF+24] analyzed the protocol cq+ in the AGM with ROM, but suggested that proving the security in AGMOS is an interesting open direction. Nitulescu, Paslis, and Ràfols [NPR26] studied their folding scheme for R1CS instances in the AGM and declared that they do not consider AGMOS for simplicity. Finally, Wagner and Zapico [WZ24] proved position-binding and code-binding of their erasure code commitment scheme in the AGM. They proposed studying code-binding in AGMOS as an interesting problem for future work.

Independently of Lipmaa et al. [LPS23], Bauer et al. [BFHK24] studied the security of the Uber-Knowledge assumption in a related model, where the adversary can leverage hashing to the group to obliviously sample group elements. Aside from the GGM and AGM(OS), Jager and Rupp [JR10] introduced the Semi-Generic Group Model for bilinear groups, where adversaries behave generically in the source groups and arbitrarily in the target group. Jaeger and Mohan [JM24] showed that, under specific conditions, proof of security in the AGM implies security in the GGM. As noted in [LPS25], this AGM-to-GGM lifting result also carries over to AGMOS.

1.4 Organization of the Paper

In Section 2, we recall the formal definitions of the Algebraic Group Model (AGM) and the AGM with Oblivious Sampling (AGMOS) (Definitions 1 and 2), alongside the Tensor Oracle Find Representation (TOFR) and Find Polynomial Representation (FPR) assumptions (Definitions 3 and 4). In Section 3, we explore the lifting of security proofs for a class of *general* AGMOS security games. Using the notation introduced by Lipmaa et al. [LPS23], we define the class of general security games in Section 3.1, encompassing both publicly and privately verifiable assumptions. Next, we describe the general structure of AGMOS security proofs and formally state and prove our main lifting theorem (Theorem 1) in Section 3.2. We then demonstrate the application of Theorem 1 by lifting the extractability property of the univariate KZG polynomial commitment scheme from the AGM to the AGMOS in Section 3.3. Subsequently, we discuss the limitations of our lifting approach in Section 3.4. Finally, we conclude Section 3 by proving the AGMOS security of the multivariate ARSDH assumption.

2 Preliminaries

For a distribution D, we write $x \leftarrow D$ to denote that the value x is sampled according to D. A finite set S in the place of D implies the uniform distribution on S. By λ, we denote the security parameter. For an algorithm $\mathcal{A}$, $y \leftarrow \mathcal{A}(x)$ denotes that given input x, $\mathcal{A}$ outputs y. By $\mathsf{poly}(\lambda)$, we denote the set $\{f(\lambda) : \exists k \in \mathbb{N},\ f(\lambda) \in O(\lambda^k)\}$. We say that a function $\varepsilon \colon \mathbb{N} \to \mathbb{R}^+$ is negligible if it approaches zero faster than any inverse polynomial. In that case, we write $\varepsilon(\lambda) \in \mathsf{negl}(\lambda)$. By $i \in [a, b]$ where $a, b \in \mathbb{Z}$, we mean that $i \in \{a, a+1, \ldots, b\}$. By PPT, we denote a probabilistic polynomial time algorithm, that is, a randomized algorithm that runs in time polynomial in the security parameter. By DPT, we denote a deterministic polynomial time algorithm, that is, a deterministic algorithm whose running time is polynomial in the security parameter.

A bilinear group generator $\mathsf{PGen}(1^\lambda)$ returns group parameters $\mathsf{gp} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, [1]_1, [1]_2, [1]_T, e)$. The groups $(\mathbb{G}_1, +)$ and $(\mathbb{G}_2, +)$ are additive cyclic groups of order p, generated by $[1]_1$ and $[1]_2$, respectively, and $(\mathbb{G}_T, \odot)$ is a multiplicative cyclic group of the same order with generator $[1]_T$. The map $e\colon \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ is a pairing, i.e., a non-degenerate efficiently computable bilinear map. Implicitly, gp contains information about the λ which was used to generate it. We use the notation $g \bullet h = e(g, h)$ for $g \in \mathbb{G}_1$ and $h \in \mathbb{G}_2$. The pairing is assumed to be Type-3, i.e., there is no efficient isomorphism between groups $\mathbb{G}_1$ and $\mathbb{G}_2$.

By $\mathbb{F}$ we denote the field $\mathbb{Z}_p$, and for indeterminates $\boldsymbol{X} = (X_1, \ldots, X_k)$,

$$\mathbb{F}^{(\leq d)}[\boldsymbol{X}] = \{f(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}] : \deg(f(\boldsymbol{X})) \leq d\},$$

where $\deg(f(\boldsymbol{X}))$ is the total degree of $f(\boldsymbol{X})$. For $m \in \mathbb{N}$ and $\mathbb{a} = (a_1, \ldots, a_m) \in \mathbb{F}^m$, we denote by $[\mathbb{a}]_\kappa = [a_1, \ldots, a_m]_\kappa$ the vector $(a_1[1]_\kappa, \ldots, a_m[1]_\kappa)$. Despite this notation, vectors are, by default, column vectors.

2.1 AGM and AGMOS

In this section, we present the formal definitions of the AGM and AGMOS.

Definition 1 (AGM [FKL18]). *A Probabilistic Polynomial Time (PPT) algorithm $\mathcal{A}$ is* algebraic *if for every group element it outputs, it also outputs its* algebraic representation *(or* explanation*) with respect to its input group elements. Symbolically,*

$$(([\mathbb{y}_1]_1, \boldsymbol{\gamma}_1), ([\mathbb{y}_2]_2, \boldsymbol{\gamma}_2)) \leftarrow \mathcal{A}(\mathsf{gp}, [\mathbb{x}_1]_1, [\mathbb{x}_2]_2),$$

where $\boldsymbol{\gamma}_1$, $\boldsymbol{\gamma}_2$ are matrices over $\mathbb{F}$ satisfying $\boldsymbol{\gamma}_1 \mathbb{x}_1 = \mathbb{y}_1$, $\boldsymbol{\gamma}_2 \mathbb{x}_2 = \mathbb{y}_2$. In the Algebraic Group Model (AGM)*, all PPT adversaries are algebraic.*

Remark 1. Given group generators $[1]_1$ and $[1]_2$, an unbounded adversary can trivially compute the discrete logarithm of any element of $\mathbb{G}_1 \cup \mathbb{G}_2$ and therefore provide its algebraic explanation. That explains why the definition focuses on PPT adversaries.

We use the definition from [FKL18], where the algebraic algorithm $\mathcal{A}$ directly outputs the algebraic explanations. We also use this approach for the AGM with oblivious sampling (AGMOS), defined below. However, a different approach was taken in [LPS23], where the AGMOS was introduced. Lipmaa et al. required that there exists an extractor which, when run on the same input and randomness as $\mathcal{A}$, outputs the algebraic explanations. As noted in [LPS23], this is a "purely cosmetic choice." More generally, extractor-based formulations are knowledge-flavored and can be subtle. In particular, the ability to break a derived computational task need not imply the ability to recover a related witness or secret (e.g., RSA inversion vs. factoring, and Dlog-based signature forgery vs. discrete logarithm). See [BV98, PV05] for canonical examples of such non-equivalences.

Although the choice described in the previous point makes Definition 1 simpler, we now have to clarify the following. Consider a security game where group elements from the adversary's output are given to another algorithm (e.g., an extractor). When analyzing the game in the AGM, the algorithm is implicitly also given the corresponding algebraic explanations. An analogous principle is used in the AGMOS.

Definition 2 (AGMOS [LPS23]). *Fix $\mathsf{gp} \leftarrow \mathsf{PGen}(1^\lambda)$. Let $\mathcal{EF}_{\mathsf{gp},\kappa}$ be a set of polynomially many functions $\mathbb{F} \to \mathbb{G}_\kappa$. Let $\mathcal{DF}_{\mathsf{gp}}$ be a family of distributions on $\mathbb{F}$. In Figure 1, we define the non-programmable $(\mathcal{EF}, \mathcal{DF})$-AGMOS oracle $\mathcal{O}$.*

*Let $\mathcal{EF} = \{\mathcal{EF}_{\mathsf{gp},\kappa}\}$ be a collection of functions. Let $\mathcal{DF} = \{\mathcal{DF}_{\mathsf{gp}}\}$ be a collection of distributions. A PPT algorithm $\mathcal{A}$ is an $(\mathcal{EF}, \mathcal{DF})$-*AGMOS adversary *for* PGen *if, given access to the oracle $\mathcal{O}$ defined above, for*

$\mathcal{O}(\kappa, E, D)$:
  If $\kappa \notin \{1,2\}$, $E \notin \mathcal{EF}_{\mathsf{gp},\kappa}$ or $D \notin \mathcal{DF}_{\mathsf{gp}}$, return $\perp$.
  Sample $s \leftarrow D$.
  Compute $[\mathbb{q}]_\kappa = E(s)$.
  Return $(s, [\mathbb{q}]_\kappa)$.

Fig. 1: The $(\mathcal{EF}, \mathcal{DF})$-AGMOS oracle $\mathcal{O}$.

every group element it outputs, it also outputs its algebraic representation *(or* explanation*) with respect to its input and oracle answers. Symbolically,*

$$([\mathbb{y}_\kappa]_\kappa, \boldsymbol{\gamma}_\kappa, \boldsymbol{\delta}_\kappa, [\mathbb{q}_\kappa]_\kappa)_{\kappa=1}^2 \leftarrow \mathcal{A}^{\mathcal{O}}(\mathsf{gp}, [\mathbb{x}_1]_1, [\mathbb{x}_2]_2),$$

where $\boldsymbol{\gamma}_\kappa$, $\boldsymbol{\delta}_\kappa$ are matrices over $\mathbb{F}$, $[\mathbb{q}_\kappa]_\kappa$ are the group elements output by $\mathcal{O}$, and it holds that

$$\boldsymbol{\gamma}_\kappa \mathbb{x}_\kappa + \boldsymbol{\delta}_\kappa \mathbb{q}_\kappa = \mathbb{y}_\kappa \quad \textit{for} \quad \kappa = 1, 2.$$

Remark 2. As mentioned in Remark 1, we use a variant of the definition where the adversary directly outputs the explanations, instead of requiring an extractor. Note that an AGMOS adversary is clearly at least as powerful as an AGM adversary.

See [LPS23] for a discussion about the choice of $\mathcal{EF}$, $\mathcal{DF}$. The authors mentioned *admissible encodings* $E \in \mathcal{EF}_{\mathsf{gp},\kappa}$ for elliptic curve groups $\mathbb{G}_\kappa$ as an example of efficient oblivious sampling. Using the definition from [LPS23], an admissible encoding is an efficiently computable function with small preimage sizes, for which the preimage of any group element in its image can be efficiently recovered. See [BF01, Ica09] for concrete examples. Not every choice of $\mathcal{EF}$, $\mathcal{DF}$ results in a meaningful model. For example, to achieve *oblivious* sampling, computing the discrete logarithm $\mathbb{q}$ of $E(s)$ from the oracle answer $(s, [\mathbb{q}]_\kappa)$ should be hard. In fact, proofs in the AGMOS actually rely on a stronger assumption TOFR, defined below.

TOFR is the first of the two key assumptions introduced in [LPS23], and it is specific to the AGMOS. As discussed in the above remark, its security depends on the parameters $\mathcal{EF}$, $\mathcal{DF}$.

Definition 3 (TOFR). *Let $\mathcal{EF}$, $\mathcal{DF}$, $\mathcal{O}$, $\mathbb{q}_\kappa$ be as in Definition 2. We say that* PGen *is* $(\mathcal{EF}, \mathcal{DF})$-$\mathsf{TOFR}$ (Tensor Oracle Find Representation) *secure if, for any PPT adversary $\mathcal{A}$,*

$$\Pr\left[\boldsymbol{v} \neq \boldsymbol{0} \ \wedge \ \boldsymbol{v}^T \begin{pmatrix} 1 \\ \mathbb{q}_1 \\ \mathbb{q}_2 \\ \mathbb{q}_1 \otimes \mathbb{q}_2 \end{pmatrix} = 0 \,\middle|\, \mathsf{gp} \leftarrow \mathsf{PGen}(1^\lambda); \boldsymbol{v} \leftarrow \mathcal{A}^{\mathcal{O}}(\mathsf{gp})\right] \in \mathsf{negl}(\lambda),$$

where $\mathbb{q}_1 \otimes \mathbb{q}_2$ denotes the Kronecker product of $\mathbb{q}_1$ and $\mathbb{q}_2$.

In other words, the probability that $\mathcal{A}$ wins the following TOFR *game* is negligible. Given group parameters gp and access to the AGMOS oracle $\mathcal{O}$, $\mathcal{A}$ must find a non-trivial linear relation between 1, $\mathbb{q}_{1i}$, $\mathbb{q}_{2j}$, $\mathbb{q}_{1i}\mathbb{q}_{2j}$, where $1 \leq i \leq \mathsf{ql}_1$, $1 \leq j \leq \mathsf{ql}_2$. Here, for $\kappa \in \{1,2\}$, ql_κ (query length) is the number of oblivious sampling queries in $\mathbb{G}_\kappa$, i.e., $\mathbb{q}_\kappa \in \mathbb{F}^{\mathsf{ql}_\kappa}$.

FPR is the second new assumption from [LPS23], which reduces to the Power Discrete Logarithm (PDL) assumption [Lip12] in the standard model. Lipmaa et al. established the reduction to PDL and then used the FPR assumption to partially automate reductions to PDL.

Definition 4 (FPR). *Let $m \geq 1$, $d_1, d_2, d_T, d_g \geq 0$ and $\boldsymbol{d} = (d_1, d_2, d_T, d_g)$. Let $\boldsymbol{X} = (X_1, \ldots, X_m)$ be indeterminates. For $\boldsymbol{x} \in \mathbb{F}^m$, let $\mathcal{O}^{\mathsf{fpr}}_{\boldsymbol{d},m,\boldsymbol{x}}(\cdot)$ be the oracle defined in Figure 2. We say that* PGen *is* $(\boldsymbol{d}, m)$-FPR (Find Polynomial Representation) secure *if, for any PPT adversary $\mathcal{A}$,*

$$\Pr\left[\begin{matrix} g(\boldsymbol{X}) \in \mathbb{F}^{(\leq d_g)}[\boldsymbol{X}] \ \wedge \\ g(\boldsymbol{X}) \neq 0 \ \wedge \ g(\boldsymbol{x}) = 0 \end{matrix} \,\middle|\, \begin{matrix} \mathsf{gp} \leftarrow \mathsf{PGen}(1^\lambda); \boldsymbol{x} \leftarrow \mathbb{F}^m; \\ g(\boldsymbol{X}) \leftarrow \mathcal{A}^{\mathcal{O}^{\mathsf{fpr}}_{\boldsymbol{d},m,\boldsymbol{x}}(\cdot)}(\mathsf{gp}) \end{matrix}\right] \in \mathsf{negl}(\lambda).$$

$\mathcal{O}^{\mathsf{fpr}}_{\boldsymbol{d},m,\boldsymbol{x}}(\kappa, f(\boldsymbol{X}))$:
If $\kappa \in \{1, 2, T\}$ and $f(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}]$ satisfies $\deg_{X_i}(f(\boldsymbol{X})) \leq d_\kappa$ for all i, then return $[f(\boldsymbol{x})]_\kappa$.
Otherwise, return $\perp$.

Fig. 2: The $(\boldsymbol{d}, m)$-FPR oracle $\mathcal{O}^{\mathsf{fpr}}_{\boldsymbol{d},m,\boldsymbol{x}}(\cdot)$.

In other words, the probability that $\mathcal{A}$ wins the following FPR *game* is negligible. The adversary is given group parameters gp and access to the FPR oracle. The oracle, on input $(\kappa, f(\boldsymbol{X}))$, checks if $f(\boldsymbol{X})$ satisfies certain degree bounds. If it does, it evaluates $f(\boldsymbol{X})$ on the secret trapdoor $\boldsymbol{x}$ and returns the corresponding $\mathbb{G}_\kappa$ element $[f(\boldsymbol{x})]_\kappa$. The adversary's task is to produce a nonzero polynomial $g(\boldsymbol{X})$ satisfying a certain degree bound and $g(\boldsymbol{x}) = 0$.

3 Lifting for general security games

Throughout this section, we distinguish two types of assumptions:

Privately verifiable assumptions: An assumption is *privately verifiable* if, in addition to the adversary's output and the public parameters $(\mathsf{gp}, [\mathbb{x}_1]_1, [\mathbb{x}_2]_2)$, the challenger requires some extra auxiliary information: $[\mathsf{aux}_1(\boldsymbol{x})]_1$ and $[\mathsf{aux}_2(\boldsymbol{x})]_2$, to determine whether the winning condition has been satisfied. Importantly, the auxiliary information is not provided to the adversary, and only used by the challenger. For instance, the $\mathsf{ARSDH}(n)$ assumption (see Section 3.5) is privately verifiable.

Publicly verifiable assumptions: An assumption is *publicly verifiable* if, from just the adversary's output and the public parameters $(\mathsf{gp}, [\mathbb{x}_1]_1, [\mathbb{x}_2]_2)$, anyone can decide whether the winning condition has been satisfied. Equivalently, an assumption is publicly verifiable if $[\mathsf{aux}_1(\boldsymbol{x})]_1$ and $[\mathsf{aux}_2(\boldsymbol{x})]_2$ are empty strings. For instance, the DLOG assumption is publicly verifiable: given the challenge $[x]_1$, one only needs to confirm that the adversary's output y satisfies $[y]_1 = [x]_1$, which is possible with no secret information required.

3.1 General security games in the AGMOS

To set the stage for our analysis in the AGMOS setting, we first describe the structure of a general security game between a challenger and an adversary. The game presented in this subsection is non-interactive, which suffices to cover a wide range of cryptographic assumptions. We use the same formalism for both privately and publicly verifiable assumptions; in the publicly verifiable case, $\mathsf{aux}_1(\boldsymbol{X})$ and $\mathsf{aux}_2(\boldsymbol{X})$ are simply empty strings. In Section 3.3, we present a publicly verifiable example by lifting extractability of the KZG polynomial commitment scheme to the AGMOS. In Section 3.5, we give a concrete application for privately verifiable security assumptions by lifting the security of the $\mathsf{ARSDH}(n)$ assumption to the AGMOS.

We begin by formally describing the general security game that we want to analyze in the AGMOS:

- The challenger runs $\mathsf{PGen}(1^\lambda)$ to obtain gp.
- The challenger samples a vector of *trapdoors* $\boldsymbol{x} \leftarrow \mathbb{F}^m$ (not known to the adversary). Let us denote by $\boldsymbol{X}$ the vector of corresponding indeterminates.
- The challenger runs $\mathsf{KPGen}(1^\lambda, \mathsf{gp})$ and :

$$(\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}), \mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X})) \in \mathbb{F}[\boldsymbol{X}]^{\mathsf{il}_1+\mathsf{il}_2+a_1+a_2}.$$

- The AGMOS adversary is given as input the group parameters gp and $([\mathbb{x}_1(\boldsymbol{x})]_1, [\mathbb{x}_2(\boldsymbol{x})]_2)$, where $\mathbb{x}_\kappa(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}]^{\mathsf{il}_\kappa}$ for $\kappa \in \{1, 2\}$, and $\mathbb{x}_{11}(\boldsymbol{X}) = \mathbb{x}_{21}(\boldsymbol{X}) = 1$.
- The adversary makes ql_κ oracle queries for each group $\mathbb{G}_\kappa$ and receives $[\mathbb{q}_\kappa]_\kappa$, $\kappa \in \{1, 2\}$.

– The adversary outputs $(([\mathbb{y}_\kappa]_\kappa, \boldsymbol{\gamma}_\kappa, \boldsymbol{\delta}_\kappa, [\mathbb{q}_\kappa]_\kappa)_{\kappa=1}^2, \boldsymbol{\alpha})$, where $\boldsymbol{\alpha} \in \mathbb{F}^r$ and

$$\mathbb{y}_\kappa \in F^{\mathsf{ol}_\kappa},\ \boldsymbol{\gamma}_\kappa \in \mathbb{F}^{\mathsf{ol}_\kappa \times \mathsf{il}_\kappa},\ \boldsymbol{\delta}_\kappa \in \mathbb{F}^{\mathsf{ol}_\kappa \times \mathsf{ql}_\kappa},\ \boldsymbol{\gamma}_\kappa \mathbb{x}_\kappa(\boldsymbol{x}) + \boldsymbol{\delta}_\kappa \mathbb{q}_\kappa = \mathbb{y}_\kappa \quad \text{for} \quad \kappa \in \{1, 2\}.$$

Here, il_κ, ol_κ, and ql_κ stand for "input length," "output length," and "query length," respectively.

– The *pairing-product verification* implicitly checks whether

$$\begin{pmatrix} \mathbb{x}_1(\boldsymbol{x}) \\ \mathbb{y}_1 \\ \mathsf{aux}_1(\boldsymbol{x}) \end{pmatrix}^T \boldsymbol{M} \begin{pmatrix} \mathbb{x}_2(\boldsymbol{x}) \\ \mathbb{y}_2 \\ \mathsf{aux}_2(\boldsymbol{x}) \end{pmatrix} = 0, \tag{1}$$

where the *verification matrix* $\boldsymbol{M} \in \mathbb{F}^{(\mathsf{il}_1 + \mathsf{ol}_1 + a_1) \times (\mathsf{il}_2 + \mathsf{ol}_2 + a_2)}$ may depend on $\boldsymbol{\alpha}$.

Next, we describe some additional useful notation for the analysis.

Explicit verification polynomial of a general security game: Let $\mathbb{Q} = (\mathbb{Q}_1, \mathbb{Q}_2)$ and $\mathbb{Y} = (\mathbb{Y}_1, \mathbb{Y}_2)$ be the corresponding vectors of indeterminates for $\mathbb{q} = (\mathbb{q}_1, \mathbb{q}_2)$ and $\mathbb{y} = (\mathbb{y}_1, \mathbb{y}_2)$, respectively. Equation (1) may be written as $V^{\mathsf{expl}}(\boldsymbol{x}, \mathbb{y}) = 0$, where

$$V^{\mathsf{expl}}(\boldsymbol{X}, \mathbb{Y}) = \begin{pmatrix} \mathbb{x}_1(\boldsymbol{X}) \\ \mathbb{Y}_1 \\ \mathsf{aux}_1(\boldsymbol{X}) \end{pmatrix}^T \boldsymbol{M} \begin{pmatrix} \mathbb{x}_2(\boldsymbol{X}) \\ \mathbb{Y}_2 \\ \mathsf{aux}_2(\boldsymbol{X}) \end{pmatrix}$$

is the *explicit verification polynomial*. In the case of publicly verifiable assumption, we have that

$$V^{\mathsf{expl}}(\boldsymbol{X}, \mathbb{Y}) = \begin{pmatrix} \mathbb{x}_1(\boldsymbol{X}) \\ \mathbb{Y}_1 \end{pmatrix}^T \boldsymbol{M} \begin{pmatrix} \mathbb{x}_2(\boldsymbol{X}) \\ \mathbb{Y}_2 \end{pmatrix}.$$

Implicit verification polynomial for privately verifiable assumptions: Let

$$\boldsymbol{M} = \begin{pmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{12} & \boldsymbol{M}_{13} \\ \boldsymbol{M}_{21} & \boldsymbol{M}_{22} & \boldsymbol{M}_{23} \\ \boldsymbol{M}_{31} & \boldsymbol{M}_{32} & \boldsymbol{M}_{33} \end{pmatrix},$$

where $\boldsymbol{M}_{11} \in \mathbb{F}^{\mathsf{il}_1 \times \mathsf{il}_2}, \boldsymbol{M}_{12} \in \mathbb{F}^{\mathsf{il}_1 \times \mathsf{ol}_2}, \boldsymbol{M}_{13} \in \mathbb{F}^{\mathsf{il}_1 \times a_2}$, etc. Let $\boldsymbol{\Gamma} = (\boldsymbol{\Gamma}_1, \boldsymbol{\Gamma}_2)$ and $\boldsymbol{\Delta} = (\boldsymbol{\Delta}_1, \boldsymbol{\Delta}_2)$ be (matrix) indeterminates corresponding to $\boldsymbol{\gamma} = (\boldsymbol{\gamma}_1, \boldsymbol{\gamma}_2)$ and $\boldsymbol{\delta} = (\boldsymbol{\delta}_1, \boldsymbol{\delta}_2)$, respectively. Let

$$\boldsymbol{P}_\kappa(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) = \begin{pmatrix} \boldsymbol{I}_{\mathsf{il}_\kappa} & \boldsymbol{0}_{\mathsf{il}_\kappa \times \mathsf{ql}_\kappa} & \boldsymbol{0}_{\mathsf{il}_\kappa \times a_\kappa} \\ \boldsymbol{\Gamma}_\kappa & \boldsymbol{\Delta}_\kappa & \boldsymbol{0}_{\mathsf{ol}_\kappa \times a_\kappa} \\ \boldsymbol{0}_{a_\kappa \times \mathsf{il}_\kappa} & \boldsymbol{0}_{a_\kappa \times \mathsf{ql}_\kappa} & \boldsymbol{I}_{a_\kappa} \end{pmatrix}$$

be the matrix so that

$$\begin{pmatrix} \mathbb{x}_\kappa(\boldsymbol{x}) \\ \mathbb{y}_\kappa \\ \mathsf{aux}_\kappa(\boldsymbol{x}) \end{pmatrix} = \boldsymbol{P}_\kappa(\boldsymbol{\gamma}, \boldsymbol{\delta}) \begin{pmatrix} \mathbb{x}_\kappa(\boldsymbol{x}) \\ \mathbb{q}_\kappa \\ \mathsf{aux}_\kappa(\boldsymbol{x}) \end{pmatrix}.$$

Define

$$\begin{aligned} \boldsymbol{N}(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) &= \boldsymbol{P}_1^T(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) \boldsymbol{M} \boldsymbol{P}_2(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) \\ &= \left(\begin{array}{c|c|c} \boldsymbol{M}_{11} + \boldsymbol{M}_{12}\boldsymbol{\Gamma}_2 + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{21} + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{22} \boldsymbol{\Gamma}_2 & (\boldsymbol{M}_{12} + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{22})\, \boldsymbol{\Delta}_2 & \boldsymbol{M}_{13} + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{23} \\ \hline \boldsymbol{\Delta}_1^T (\boldsymbol{M}_{21} + \boldsymbol{M}_{22} \boldsymbol{\Gamma}_2) & \boldsymbol{\Delta}_1^T \boldsymbol{M}_{22} \boldsymbol{\Delta}_2 & \boldsymbol{\Delta}_1^T \boldsymbol{M}_{23} \\ \hline \boldsymbol{M}_{31} + \boldsymbol{M}_{32} \boldsymbol{\Gamma}_2 & \boldsymbol{M}_{32} \boldsymbol{\Delta}_2 & \boldsymbol{M}_{33} \end{array} \right). \end{aligned}$$

Equation (1) may now be written as $V(\boldsymbol{x}, \mathbb{q}) = 0$, where

$$\begin{aligned} V(\boldsymbol{X}, \mathbb{Q}) &= V^{\mathsf{expl}}(\boldsymbol{X}, \boldsymbol{\gamma}_1 \mathbb{x}_1(\boldsymbol{X}) + \boldsymbol{\delta}_1 \mathbb{Q}_1, \boldsymbol{\gamma}_2 \mathbb{x}_2(\boldsymbol{X}) + \boldsymbol{\delta}_2 \mathbb{Q}_2) \\ &= \begin{pmatrix} \mathbb{x}_1(\boldsymbol{X}) \\ \mathbb{Q}_1 \\ \mathsf{aux}_1(\boldsymbol{X}) \end{pmatrix}^T \boldsymbol{N}(\boldsymbol{\gamma}, \boldsymbol{\delta}) \begin{pmatrix} \mathbb{x}_2(\boldsymbol{X}) \\ \mathbb{Q}_2 \\ \mathsf{aux}_2(\boldsymbol{X}) \end{pmatrix} \end{aligned}$$

is the *implicit verification polynomial*.

In the case of a publicly verifiable assumption, we have that:

$$\boldsymbol{M} = \begin{pmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{12} \\ \boldsymbol{M}_{21} & \boldsymbol{M}_{22} \end{pmatrix} \text{ and } \boldsymbol{P}_\kappa(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) = \begin{pmatrix} \boldsymbol{I}_{\mathsf{il}_\kappa} & \boldsymbol{0}_{\mathsf{il}_\kappa \times \mathsf{ql}_\kappa} \\ \boldsymbol{\Gamma}_\kappa & \boldsymbol{\Delta}_\kappa \end{pmatrix}.$$

The polynomial $V(\boldsymbol{X}, \mathbb{Q})$ is changed into:

$$V(\boldsymbol{X}, \mathbb{Q}) = \begin{pmatrix} \mathbb{x}_1(\boldsymbol{X}) \\ \mathbb{Q}_1 \end{pmatrix}^T \boldsymbol{N}(\boldsymbol{\gamma}, \boldsymbol{\delta}) \begin{pmatrix} \mathbb{x}_2(\boldsymbol{X}) \\ \mathbb{Q}_2 \end{pmatrix},$$

where the matrix $\boldsymbol{N}(\boldsymbol{\gamma}, \boldsymbol{\delta})$ is of the following form:

$$\boldsymbol{N}(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) = \left(\begin{array}{c|c} \boldsymbol{M}_{11} + \boldsymbol{M}_{12}\, \boldsymbol{\Gamma}_2 + \boldsymbol{\Gamma}_1^T\, \boldsymbol{M}_{21} + \boldsymbol{\Gamma}_1^T\, \boldsymbol{M}_{22}\, \boldsymbol{\Gamma}_2 & \left(\boldsymbol{M}_{12} + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{22}\right) \boldsymbol{\Delta}_2 \\ \hline \boldsymbol{\Delta}_1^T \left(\boldsymbol{M}_{21} + \boldsymbol{M}_{22}\, \boldsymbol{\Gamma}_2\right) & \boldsymbol{\Delta}_1^T\, \boldsymbol{M}_{22}\, \boldsymbol{\Delta}_2 \end{array}\right).$$

Separating the $\mathbb{Q}$ *terms:* We decompose the polynomial $V(\boldsymbol{X}, \mathbb{Q})$ as $V(\boldsymbol{X}, \mathbb{Q}) = V^h(\boldsymbol{X}) + V^t(\boldsymbol{X}, \mathbb{Q})$, where $V^h(\boldsymbol{X})$ collects all terms that depend only on $\boldsymbol{X}$ or are constant, and $V^t(\boldsymbol{X}, \mathbb{Q})$ collects all terms that depend either on the $\mathbb{Q}_{\kappa i}$ variables alone or jointly on both $\boldsymbol{X}$ and the $\mathbb{Q}_{\kappa i}$ variables.

Then the $V^h(\boldsymbol{X})$ part of the implicit verification polynomial is

$$\begin{aligned} V^h(\boldsymbol{X}) = {} & \mathbb{x}_1^T(\boldsymbol{X})(\boldsymbol{M}_{11} + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{21} + \boldsymbol{M}_{12}\boldsymbol{\Gamma}_2 + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{22}\boldsymbol{\Gamma}_2)\mathbb{x}_2(\boldsymbol{X}) + \\ & + \mathbb{x}_1^T(\boldsymbol{X})(\boldsymbol{M}_{13} + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{23})\mathsf{aux}_2(\boldsymbol{X}) + \mathsf{aux}_1^T(\boldsymbol{X})(\boldsymbol{M}_{31} + \boldsymbol{M}_{32}\boldsymbol{\Gamma}_2)\mathbb{x}_2(\boldsymbol{X}) + \\ & + \mathsf{aux}_1^T(\boldsymbol{X})\boldsymbol{M}_{33}\mathsf{aux}_2(\boldsymbol{X}) \end{aligned} \tag{2}$$

and the $V^t(\boldsymbol{X}, \mathbb{Q})$ part of the implicit verification polynomial is

$$V^t(\boldsymbol{X}, \mathbb{Q}) = \begin{pmatrix} \mathbb{x}_1(\boldsymbol{X}) \\ \mathbb{Q}_1 \\ \mathsf{aux}_1(\boldsymbol{X}) \end{pmatrix}^T \boldsymbol{f_M}(\boldsymbol{\gamma}, \boldsymbol{\delta}) \begin{pmatrix} \mathbb{x}_2(\boldsymbol{X}) \\ \mathbb{Q}_2 \\ \mathsf{aux}_2(\boldsymbol{X}) \end{pmatrix}, \tag{3}$$

where

$$\boldsymbol{f_M}(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) = \begin{pmatrix} \boldsymbol{0}_{\mathsf{il}_1 \times \mathsf{il}_2} & (\boldsymbol{M}_{12} + \boldsymbol{\Gamma}_1^T \boldsymbol{M}_{22})\boldsymbol{\Delta}_2 & \boldsymbol{0}_{\mathsf{il}_1 \times a_2} \\ \boldsymbol{\Delta}_1^T(\boldsymbol{M}_{21} + \boldsymbol{M}_{22}\boldsymbol{\Gamma}_2) & \boldsymbol{\Delta}_1^T \boldsymbol{M}_{22}\boldsymbol{\Delta}_2 & \boldsymbol{\Delta}_1^T \boldsymbol{M}_{23} \\ \boldsymbol{0}_{a_1 \times \mathsf{il}_2} & \boldsymbol{M}_{32}\boldsymbol{\Delta}_2 & \boldsymbol{0}_{a_1 \times a_2} \end{pmatrix} \tag{4}$$

is equal to $\boldsymbol{N}(\boldsymbol{\Gamma}, \boldsymbol{\Delta})$ except that its corner submatrices are $\boldsymbol{0}$. To emphasize how $V^t(\boldsymbol{X}, \mathbb{Q})$ is constructed, let us denote

$$V^t_{\mathbb{x}, \mathsf{aux}, \boldsymbol{M}, \boldsymbol{\Gamma}, \boldsymbol{\Delta}}(\boldsymbol{X}, \mathbb{Q}) = \begin{pmatrix} \mathbb{x}_1(\boldsymbol{X}) \\ \mathbb{Q}_1 \\ \mathsf{aux}_1(\boldsymbol{X}) \end{pmatrix}^T \boldsymbol{f_M}(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) \begin{pmatrix} \mathbb{x}_2(\boldsymbol{X}) \\ \mathbb{Q}_2 \\ \mathsf{aux}_2(\boldsymbol{X}) \end{pmatrix}, \tag{5}$$

where $\mathbb{x} = (\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}))$ and $\mathsf{aux} = (\mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X}))$.

When considering publicly verifiable assumptions, the polynomial $V^t(\boldsymbol{X}, \mathbb{Q})$ is the following

$$V^t_{\mathbb{x}, \boldsymbol{M}, \boldsymbol{\Gamma}, \boldsymbol{\Delta}}(\boldsymbol{X}, \mathbb{Q}) = \begin{pmatrix} \mathbb{x}_1(\boldsymbol{X}) \\ \mathbb{Q}_1 \end{pmatrix}^T \boldsymbol{f_M}(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) \begin{pmatrix} \mathbb{x}_2(\boldsymbol{X}) \\ \mathbb{Q}_2 \end{pmatrix},$$

where

$$f_{M}(\Gamma, \Delta) = \begin{pmatrix} \mathbf{0}_{\mathsf{il}_1 \times \mathsf{il}_2} & (M_{12} + \Gamma_1^T M_{22})\Delta_2 \\ \Delta_1^T (M_{21} + M_{22}\Gamma_2) & \Delta_1^T M_{22} \Delta_2 \end{pmatrix}.$$

In the next subsection, we show that the structure of the $V^t(\boldsymbol{X}, \mathbb{Q})$ is the key difference between the AGM proofs of assumptions which can be lifted to the AGMOS and those which cannot.

3.2 Our Lifting Theorem

We start with an overview of the general AGMOS proof strategy from [LPS23]. We focus on the class of security games described in the previous section, although their proof strategy is more general. The AGMOS adversary's output is accepted if the implicit verification polynomial $V(\boldsymbol{X}, \mathbb{Q})$ satisfies $V(\boldsymbol{x}, \mathbb{q}) = 0$, where $\boldsymbol{x}$ are the trapdoors and $[\mathbb{q}_{\kappa}]_{\kappa}$ the AGMOS oracle answers. One then analyzes four cases:

Case A**:** $V(\boldsymbol{X}, \mathbb{Q}) = 0$**.** Typically, this case implies that the adversary failed. Thus, it does not require treatment in the proof of security.
Case $\mathsf{X.1}$**:** $V(\boldsymbol{X}, \mathbb{Q}) \neq 0$**, but** $V^t(\boldsymbol{X}, \mathbb{Q}) = 0$**.** One constructs a reduction to FPR.
Case $\mathsf{X.2}$**:** $V^t(\boldsymbol{X}, \mathbb{Q}) \neq 0$**, but** $V^t(\boldsymbol{x}, \mathbb{Q}) = 0$**.** One constructs a different reduction to FPR.
Case Q**:** $V^t(\boldsymbol{x}, \mathbb{Q}) \neq 0$**.** One constructs a reduction to TOFR.

In the AGM, the verification polynomial does not include variables $\mathbb{Q}$, so we have $V^t(\boldsymbol{X}, \mathbb{Q}) = 0$ and only cases A and $\mathsf{X.1}$ must be dealt with.

To establish the lifting theorem, we prove that, under several assumptions, an AGMOS adversary $\mathcal{A}$ which passes a verification of the form (1) is essentially an AGM adversary. This means that, with overwhelming probability, $\mathcal{A}$ outputs $\boldsymbol{\delta} = \mathbf{0}$, i.e., although $\mathcal{A}$ can query the AGMOS oracle $\mathcal{O}$, it does not use its answers to explain its output. We make the implicit assumption that the functions from $\mathcal{EF}$ are efficiently computable and the distributions from $\mathcal{DF}$ efficiently sampleable, so that $\mathcal{O}$ can be efficiently simulated. Our result allows one to lift some AGM proofs to the AGMOS instead of repeating the whole AGM proof just to add the AGMOS-specific parts.

Motivating our lifting condition. First, consider cases A and $\mathsf{X.1}$, where $V^t(\boldsymbol{X}, \mathbb{Q}) = 0$. Note that we cannot establish a lifting theorem for arbitrary security games, because the AGMOS adversary can pass the verification with $\boldsymbol{\delta} \neq \mathbf{0}$ in some cases. Indeed, Lipmaa et al. used this observation to separate the AGM from the AGMOS (see [LPS23, Section 6] and our discussion in Section 3.4). They showed that, if the DLOG assumption holds, some assumptions that are secure in the AGM are not secure in the AGMOS. To achieve lifting from AGM to the AGMOS, we must therefore add some structural condition on the security game. We chose the natural *lifting condition* that $V^t(\boldsymbol{X}, \mathbb{Q}) = 0$ implies $\boldsymbol{\delta} = \mathbf{0}$. For our lifting theorem to apply, the lifting condition must hold for the specific assumption we wish to lift from AGM to AGMOS. Thus, one has to verify the lifting condition before invoking the lifting theorem. However, the verification is straightforward. By Equations (4) and (5), the equation $V^t_{\mathbb{x},\mathsf{aux},\boldsymbol{M},\boldsymbol{\Gamma},\boldsymbol{\Delta}}(\boldsymbol{X}, \mathbb{Q}) = 0$ can be formulated as a system of polynomial equations in $\boldsymbol{M}$, $\boldsymbol{\Gamma}$ and $\boldsymbol{\Delta}$, which are linear or quadratic in $\boldsymbol{\Delta}$. Therefore, to verify the lifting condition, it is sufficient to construct the system of equations and verify that it admits only the trivial solution (see Section 3.3 for an example).

Handling the AGMOS specific remaining cases. Second, consider cases $\mathsf{X.2}$ and Q, which are specific to the AGMOS. As mentioned in [LPS23], the assumptions FPR and TOFR are almost tautological for these respective two cases.

- In case $\mathsf{X.2}$, consider $V^t(\boldsymbol{X}, \mathbb{Q})$ as a polynomial in the indeterminates $\mathbb{Q}$ over $\mathbb{F}[\boldsymbol{X}]$. Since $V^t(\boldsymbol{X}, \mathbb{Q}) \neq 0$, one of its coefficients—let us denote it by $g(\boldsymbol{X})$—is nonzero. Since $V^t(\boldsymbol{x}, \mathbb{Q}) = 0$, we have $g(\boldsymbol{x}) = 0$. Being able to construct such a polynomial contradicts the FPR assumption.
- In case Q, the coefficients of $V^h(\boldsymbol{x}) + V^t(\boldsymbol{x}, \mathbb{Q}) \in \mathbb{F}[\mathbb{Q}]$ correspond to the components of $\boldsymbol{v}$ from Definition 3, allowing us to break the TOFR assumption. We deal with these two cases in Theorem 1.

Capturing constructions of adversary's inputs and adaptive verification equations. In a security game, the trapdoor can be used to produce many group elements that are given as input to the adversary, e.g., discrete logarithm commitments to q powers of the trapdoor in the q-DLog assumption. Moreover, the matrix specifying the verification equation often depends on the adversaries' outputs. To formally capture the above possibilities, we introduce DPT algorithms KPGen and MGen.

The algorithm KPGen generates the polynomials $\mathbb{x}_\kappa(\boldsymbol{X})$, which are used to compute the adversary's input. If the assumption under consideration is privately verifiable, KPGen additionally outputs the auxiliary polynomials $\mathsf{aux}_\kappa(\boldsymbol{X})$ used to verify the adversary's output. As an example, consider the extractability game for KZG (see Section 3.3), where KPGen simply outputs the powers X^i of the indeterminate X.

The algorithm MGen generates the verification matrix $\boldsymbol{M}$, which may depend on field elements $\boldsymbol{\alpha}$ returned by the adversary. Continuing with the KZG example, the verification depends on the evaluation point α and the claimed evaluation η, and, correspondingly, the verification matrix also depends on $\boldsymbol{\alpha}$.

Lifting theorem for general security games. Now, we state and prove our lifting theorem for general security games. First, we formally define the lifting condition.

Definition 5 (Lifting condition). *Let $m \in \mathbb{N}$, $r, \mathsf{il}_\kappa, \mathsf{ol}_\kappa, a_\kappa \in \mathsf{poly}(\lambda)$ for $\kappa \in \{1,2\}$. Let $\boldsymbol{d} = (d_1, d_2, d_T, d_g)$. Let $\boldsymbol{X} = (X_1, \ldots, X_m)$ be indeterminates. Let MGen and KPGen be DPT algorithms such that, for all $\lambda \in \mathbb{N}$ and $\boldsymbol{\alpha} \in \mathbb{F}^r$,*

$$\mathsf{MGen}(1^\lambda, \mathsf{gp}, \boldsymbol{\alpha}) \to \boldsymbol{M} \in \mathbb{F}^{(\mathsf{il}_1+\mathsf{ol}_1+a_1)\times(\mathsf{il}_2+\mathsf{ol}_2+a_2)},$$
$$\mathsf{KPGen}(1^\lambda, \mathsf{gp}) \to (\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}), \mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X})) \in \mathbb{F}[\boldsymbol{X}]^{\mathsf{il}_1+\mathsf{il}_2+a_1+a_2},$$

with

$$\mathbb{x}_{11}(\boldsymbol{X}) = \mathbb{x}_{21}(\boldsymbol{X}) = 1,\ \mathbb{x}_{\kappa i}(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}], \deg_{X_j}(\mathbb{x}_{\kappa i}(\boldsymbol{X})) \le d_\kappa \text{ and } \deg_{X_j}(\mathsf{aux}_{\kappa i}(\boldsymbol{X})) \le d_\kappa \tag{6}$$

for all i, j, κ, and $d_g \ge \max(md_1, md_2)$.

We say that the lifting condition *holds for the security game Γ from Section 3.1 defined by the above parameters if, for all $\lambda \in \mathbb{N}$, $\mathsf{ql}_\kappa \in \mathsf{poly}(\lambda)$, $\boldsymbol{\gamma}_\kappa \in \mathbb{F}^{\mathsf{ol}_\kappa \times \mathsf{il}_\kappa}$, $\boldsymbol{\delta}_\kappa \in \mathbb{F}^{\mathsf{ol}_\kappa \times \mathsf{ql}_\kappa}$ for $\kappa \in \{1,2\}$ and $\boldsymbol{\alpha} \in \mathbb{F}^r$:*

$$\begin{aligned} \textit{if} \quad & \mathsf{MGen}(1^\lambda, \mathsf{gp}, \boldsymbol{\alpha}) \to \boldsymbol{M}, \\ & \mathsf{KPGen}(1^\lambda, \mathsf{gp}) \to (\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}), \mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X})) \\ \textit{and} \quad & V^t_{\mathbb{x},\mathsf{aux},\boldsymbol{M},\boldsymbol{\gamma},\boldsymbol{\delta}}(\boldsymbol{X}, \mathbb{Q}) = 0, \\ \textit{then} \quad & \boldsymbol{\delta} = \boldsymbol{0}. \end{aligned}$$

Theorem 1 (Lifting theorem). *Let m, r, il_κ, ol_κ, a_κ, $\mathcal{EF}$, $\mathcal{DF}$, $\boldsymbol{d}$, $\boldsymbol{X}$, MGen, KPGen be parameters as in Definition 5 specifying an AGMOS security game Γ from Section 3.1 satisfying the lifting condition. If the $(\boldsymbol{d}, m)$-FPR and $(\mathcal{EF}, \mathcal{DF})$-$\mathsf{TOFR}$ assumptions hold then, for every PPT $(\mathcal{EF}, \mathcal{DF})$-AGMOS adversary $\mathcal{A}$, there exists a PPT AGM adversary $\mathcal{B}$ such that*

$$\Pr\big[\mathcal{A} \textit{ wins } \Gamma\big] \le \Pr\big[\mathcal{B} \textit{ wins the AGM variant of } \Gamma\big] + \mathsf{negl}(\lambda).$$

Proof. Fix a PPT $(\mathcal{EF}, \mathcal{DF})$-AGMOS adversary $\mathcal{A}$ for Γ. Let Win be the event that $\mathcal{A}$ passes the pairing-product verification check Equation (1). Let Bad be the event that $\boldsymbol{\delta} \ne \boldsymbol{0}$ in $\mathcal{A}$'s output. We decompose the winning probability as

$$\Pr[\mathsf{Win}] = \Pr[\mathsf{Win} \wedge \neg\mathsf{Bad}] + \Pr[\mathsf{Win} \wedge \mathsf{Bad}].$$

First, we bound the term $\Pr[\mathsf{Win} \wedge \mathsf{Bad}]$. By the lifting condition (Definition 5), we have the implication $V^t_{\mathbb{x},\mathsf{aux},\boldsymbol{M},\boldsymbol{\gamma},\boldsymbol{\delta}}(\boldsymbol{X}, \mathbb{Q}) = 0 \Rightarrow \boldsymbol{\delta} = \boldsymbol{0}$. Taking the contrapositive, the event Bad (i.e., $\boldsymbol{\delta} \ne \boldsymbol{0}$) implies that the polynomial $V^t_{\mathbb{x},\mathsf{aux},\boldsymbol{M},\boldsymbol{\gamma},\boldsymbol{\delta}}(\boldsymbol{X}, \mathbb{Q})$ is not the zero polynomial. Consequently,

$$\Pr[\mathsf{Win} \wedge \mathsf{Bad}] \le \Pr\big[\mathsf{Win} \wedge (V^t(\boldsymbol{X}, \mathbb{Q}) \ne 0)\big].$$

We can write the probability on the right-hand side explicitly as:

$$\Pr\left[\begin{array}{l|l} \mathbb{y}_1 \in \mathbb{F}^{\mathsf{ol}_1} \ \wedge \ \mathbb{y}_2 \in \mathbb{F}^{\mathsf{ol}_2} \ \wedge \ \boldsymbol{\alpha} \in \mathbb{F}^r \ \wedge & \mathsf{gp} \leftarrow \mathsf{PGen}(1^\lambda); \boldsymbol{x} \leftarrow \mathbb{F}^m; \\ \begin{pmatrix} \mathbb{x}_1(\boldsymbol{x}) \\ \mathbb{y}_1 \\ \mathsf{aux}_1(\boldsymbol{x}) \end{pmatrix}^T \boldsymbol{M} \begin{pmatrix} \mathbb{x}_2(\boldsymbol{x}) \\ \mathbb{y}_2 \\ \mathsf{aux}_2(\boldsymbol{x}) \end{pmatrix} = 0 \ \wedge & \begin{array}{l} (\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}), \mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X})) \leftarrow \mathsf{KPGen}(1^\lambda, \mathsf{gp}); \\ (([\mathbb{y}_\kappa]_\kappa, \boldsymbol{\gamma}_\kappa, \boldsymbol{\delta}_\kappa, [\mathbb{q}_\kappa]_\kappa)_{\kappa=1}^2, \boldsymbol{\alpha}) \leftarrow \\ \quad \leftarrow \mathcal{A}^{\mathcal{O}}(\mathsf{gp}, [\mathbb{x}_1(\boldsymbol{x})]_1, [\mathbb{x}_2(\boldsymbol{x})]_2); \end{array} \\ V^t(\boldsymbol{X}, \mathbb{Q}) \neq 0 & \boldsymbol{M} \leftarrow \mathsf{MGen}(1^\lambda, \mathsf{gp}, \boldsymbol{\alpha}) \end{array}\right]. \tag{7}$$

We bound the probability in Equation (7) in Lemma 1 below, under the FPR and TOFR assumptions. Thus, we get $\Pr[\mathsf{Win} \wedge \mathsf{Bad}] \in \mathsf{negl}(\lambda)$.

Next, we consider the term $\Pr[\mathsf{Win} \wedge \neg\mathsf{Bad}]$. We define an AGM adversary $\mathcal{B}$. On input $(\mathsf{gp}, [\mathbb{x}_1(\boldsymbol{x})]_1, [\mathbb{x}_2(\boldsymbol{x})]_2)$, $\mathcal{B}$ runs $\mathcal{A}$ and answers its oracle calls by locally simulating the $(\mathcal{EF}, \mathcal{DF})$-AGMOS oracle $\mathcal{O}$ from Figure 1. This simulation is efficient because the functions in $\mathcal{EF}$ are efficiently computable and the distributions in $\mathcal{DF}$ are efficiently sampleable. When $\mathcal{A}$ terminates, $\mathcal{B}$ checks whether $\boldsymbol{\delta} = \mathbf{0}$. If $\boldsymbol{\delta} = \mathbf{0}$, then $\mathcal{B}$ forwards $\mathcal{A}$'s claimed output group elements and the explanation matrices $\boldsymbol{\gamma}_\kappa$ to the AGM challenger. Otherwise, $\mathcal{B}$ outputs $\perp$. Whenever $\mathsf{Win} \wedge \neg\mathsf{Bad}$ occurs, $\mathcal{A}$ wins the game and uses only the challenger-provided input basis (since $\boldsymbol{\delta} = \mathbf{0}$). Thus, $\mathcal{B}$ wins the AGM variant of the game exactly when $\mathcal{A}$ wins and $\boldsymbol{\delta} = \mathbf{0}$. Hence,

$$\Pr[\mathsf{Win} \wedge \neg\mathsf{Bad}] \leq \Pr\big[\mathcal{B} \text{ wins the AGM variant of the game}\big].$$

Combining the bounds concludes the proof. □

Lemma 1 (Security for the AGMOS-specific cases). *Let $m \in \mathbb{N}$, $r, \mathsf{il}_\kappa, \mathsf{ol}_\kappa, a_\kappa \in \mathsf{poly}(\lambda)$ for $\kappa \in \{1,2\}$. Let $\mathcal{A}$ be a PPT $(\mathcal{EF}, \mathcal{DF})$-AGMOS adversary. Let $\boldsymbol{d} = (d_1, d_2, d_T, d_g)$. Let $\boldsymbol{X} = (X_1, \ldots, X_m)$ be indeterminates. Let* MGen *and* KPGen *be DPT algorithms such that, for all $\lambda \in \mathbb{N}$ and $\boldsymbol{\alpha} \in \mathbb{F}^r$,*

$$\begin{aligned} &\mathsf{MGen}(1^\lambda, \mathsf{gp}, \boldsymbol{\alpha}) \to \boldsymbol{M} \in \mathbb{F}^{(\mathsf{il}_1 + \mathsf{ol}_1 + a_1) \times (\mathsf{il}_2 + \mathsf{ol}_2 + a_2)}, \\ &\mathsf{KPGen}(1^\lambda, \mathsf{gp}) \to (\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}), \mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X})) \in \mathbb{F}[\boldsymbol{X}]^{\mathsf{il}_1 + \mathsf{il}_2 + a_1 + a_2}, \end{aligned}$$

with

$$\mathbb{x}_{11}(\boldsymbol{X}) = \mathbb{x}_{21}(\boldsymbol{X}) = 1, \ \mathbb{x}_{\kappa i}(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}], \deg_{X_j}(\mathbb{x}_{\kappa i}(\boldsymbol{X})) \leq d_\kappa \text{ and } \deg_{X_j}(\mathsf{aux}_{\kappa i}(\boldsymbol{X})) \leq d_\kappa$$

*for all i, j, κ. Suppose that $d_g \geq \max(md_1, md_2)$. If the $(\boldsymbol{d}, m)$-*FPR *and $(\mathcal{EF}, \mathcal{DF})$-*TOFR *assumptions hold, then*

$$\Pr\left[\begin{array}{l|l} \mathbb{y}_1 \in \mathbb{F}^{\mathsf{ol}_1} \ \wedge \ \mathbb{y}_2 \in \mathbb{F}^{\mathsf{ol}_2} \ \wedge \ \boldsymbol{\alpha} \in \mathbb{F}^r \ \wedge & \mathsf{gp} \leftarrow \mathsf{PGen}(1^\lambda); \boldsymbol{x} \leftarrow \mathbb{F}^m; \\ \begin{pmatrix} \mathbb{x}_1(\boldsymbol{x}) \\ \mathbb{y}_1 \\ \mathsf{aux}_1(\boldsymbol{x}) \end{pmatrix}^T \boldsymbol{M} \begin{pmatrix} \mathbb{x}_2(\boldsymbol{x}) \\ \mathbb{y}_2 \\ \mathsf{aux}_2(\boldsymbol{x}) \end{pmatrix} = 0 \ \wedge & \begin{array}{l} (\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}), \mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X})) \leftarrow \mathsf{KPGen}(1^\lambda, \mathsf{gp}); \\ (([\mathbb{y}_\kappa]_\kappa, \boldsymbol{\gamma}_\kappa, \boldsymbol{\delta}_\kappa, [\mathbb{q}_\kappa]_\kappa)_{\kappa=1}^2, \boldsymbol{\alpha}) \leftarrow \\ \quad \leftarrow \mathcal{A}^{\mathcal{O}}(\mathsf{gp}, [\mathbb{x}_1(\boldsymbol{x})]_1, [\mathbb{x}_2(\boldsymbol{x})]_2); \end{array} \\ V^t(\boldsymbol{X}, \mathbb{Q}) \neq 0 & \boldsymbol{M} \leftarrow \mathsf{MGen}(1^\lambda, \mathsf{gp}, \boldsymbol{\alpha}) \end{array}\right] \in \mathsf{negl}(\lambda),$$

where $\boldsymbol{\alpha}$ are the field elements in $\mathcal{A}$'s output and $V^t(\boldsymbol{X}, \mathbb{Q})$ is as in Equation (3).

Proof. Suppose that $\mathcal{A}$ passes the pairing-product verification check and $V^t(\boldsymbol{X}, \mathbb{Q}) \neq 0$. Using Equations (3) and (4), we derive

$$\begin{aligned} V^t(\boldsymbol{X}, \mathbb{Q}) &= \mathbb{x}_1^T(\boldsymbol{X})(\boldsymbol{M}_{12}\boldsymbol{\delta}_2 + \boldsymbol{\gamma}_1^T \boldsymbol{M}_{22}\boldsymbol{\delta}_2)\mathbb{Q}_2 + \mathbb{Q}_1^T \boldsymbol{\delta}_1^T (\boldsymbol{M}_{21} + \boldsymbol{M}_{22}\boldsymbol{\gamma}_2)\mathbb{x}_2(\boldsymbol{X}) \\ &\quad + \mathbb{Q}_1^T \boldsymbol{\delta}_1^T \boldsymbol{M}_{22}\boldsymbol{\delta}_2 \mathbb{Q}_2 + \mathbb{Q}_1^T \boldsymbol{\delta}_1^T \boldsymbol{M}_{23} \mathsf{aux}_2(\boldsymbol{X}) + \mathsf{aux}_1^T(\boldsymbol{X}) \boldsymbol{M}_{32}\boldsymbol{\delta}_2 \mathbb{Q}_2 \\ &= \begin{pmatrix} \boldsymbol{\delta}_1^T (\boldsymbol{M}_{21} + \boldsymbol{M}_{22}\boldsymbol{\gamma}_2)\mathbb{x}_2(\boldsymbol{X}) + \boldsymbol{\delta}_1^T \boldsymbol{M}_{23} \mathsf{aux}_2(\boldsymbol{X}) \\ \boldsymbol{\delta}_2^T (\boldsymbol{M}_{12}^T + \boldsymbol{M}_{22}^T \boldsymbol{\gamma}_1)\mathbb{x}_1(\boldsymbol{X}) + \boldsymbol{\delta}_2^T \boldsymbol{M}_{32}^T \mathsf{aux}_1(\boldsymbol{X}) \\ \mathrm{vec}(\boldsymbol{\delta}_2^T \boldsymbol{M}_{22}^T \boldsymbol{\delta}_1) \end{pmatrix}^T \begin{pmatrix} \mathbb{Q}_1 \\ \mathbb{Q}_2 \\ \mathbb{Q}_1 \otimes \mathbb{Q}_2 \end{pmatrix}, \end{aligned} \tag{8}$$

$\mathcal{B}_{\mathsf{fpr}}(\mathsf{gp})$:

Run $\mathsf{KPGen}(1^\lambda, \mathsf{gp}) \to (\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}), \mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X}))$.

Query $\mathcal{O}^{\mathsf{fpr}}_{\boldsymbol{d},m,\boldsymbol{x}}(\cdot)$ to get $[\mathbb{x}_1(\boldsymbol{x})]_1$ and $[\mathbb{x}_2(\boldsymbol{x})]_2$.

Run $\mathcal{A}^{\mathcal{O}}(\mathsf{gp}, [\mathbb{x}_1(\boldsymbol{x})]_1, [\mathbb{x}_2(\boldsymbol{x})]_2) \to (([\mathbb{y}_\kappa]_\kappa, \boldsymbol{\gamma}_\kappa, \boldsymbol{\delta}_\kappa, [\mathbb{q}_\kappa]_\kappa)_{\kappa=1}^2, \boldsymbol{\alpha})$.

Run $\mathsf{MGen}(1^\lambda, \mathsf{gp}, \boldsymbol{\alpha}) \to \boldsymbol{M}$.

Find a nonzero component $g(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}]$ of

$$\begin{pmatrix} \boldsymbol{\delta}_1^T(\boldsymbol{M}_{21} + \boldsymbol{M}_{22}\boldsymbol{\gamma}_2)\mathbb{x}_2(\boldsymbol{X}) + \boldsymbol{\delta}_1^T\boldsymbol{M}_{23}\mathsf{aux}_2(\boldsymbol{X}) \\ \boldsymbol{\delta}_2^T(\boldsymbol{M}_{12}^T + \boldsymbol{M}_{22}^T\boldsymbol{\gamma}_1)\mathbb{x}_1(\boldsymbol{X}) + \boldsymbol{\delta}_2^T\boldsymbol{M}_{32}^T\mathsf{aux}_1(\boldsymbol{X}) \end{pmatrix}$$

and return it.

Fig. 3: The adversary $\mathcal{B}_{\mathsf{fpr}}$ for the $(\boldsymbol{d}, m)$-FPR game in Lemma 1.

$\mathcal{B}_{\mathsf{tofr}}(\mathsf{gp})$:

Run $\mathsf{KPGen}(1^\lambda, \mathsf{gp}) \to (\mathbb{x}_1(\boldsymbol{X}), \mathbb{x}_2(\boldsymbol{X}), \mathsf{aux}_1(\boldsymbol{X}), \mathsf{aux}_2(\boldsymbol{X}))$.

Sample $\boldsymbol{x} \leftarrow \mathbb{F}^m$.

Run $\mathcal{A}^{\mathcal{O}}(\mathsf{gp}, [\mathbb{x}_1(\boldsymbol{x})]_1, [\mathbb{x}_2(\boldsymbol{x})]_2) \to (([\mathbb{y}_\kappa]_\kappa, \boldsymbol{\gamma}_\kappa, \boldsymbol{\delta}_\kappa, [\mathbb{q}_\kappa]_\kappa)_{\kappa=1}^2, \boldsymbol{\alpha})$.

Run $\mathsf{MGen}(1^\lambda, \mathsf{gp}, \boldsymbol{\alpha}) \to \boldsymbol{M}$.

Compute $V^h(\boldsymbol{X})$ from $\mathbb{x}_\kappa(\boldsymbol{X})$, $\mathsf{aux}_\kappa(\boldsymbol{X})$, $\boldsymbol{\gamma}$ and $\boldsymbol{M}$ (see Equation (2)).

Return

$$\boldsymbol{v} = \begin{pmatrix} V^h(\boldsymbol{x}) \\ \boldsymbol{\delta}_1^T(\boldsymbol{M}_{21} + \boldsymbol{M}_{22}\boldsymbol{\gamma}_2)\mathbb{x}_2(\boldsymbol{x}) + \boldsymbol{\delta}_1^T\boldsymbol{M}_{23}\mathsf{aux}_2(\boldsymbol{x}) \\ \boldsymbol{\delta}_2^T(\boldsymbol{M}_{12}^T + \boldsymbol{M}_{22}^T\boldsymbol{\gamma}_1)\mathbb{x}_1(\boldsymbol{x}) + \boldsymbol{\delta}_2^T\boldsymbol{M}_{32}^T\mathsf{aux}_1(\boldsymbol{x}) \\ \mathrm{vec}(\boldsymbol{\delta}_2^T\boldsymbol{M}_{22}^T\boldsymbol{\delta}_1) \end{pmatrix}.$$

Fig. 4: The adversary $\mathcal{B}_{\mathsf{tofr}}$ for the $(\mathcal{EF}, \mathcal{DF})$-$\mathsf{TOFR}$ game in Lemma 1.

where the operator vec denotes vectorization.

Case $\mathsf{X.2}$**:** $V^t(\boldsymbol{X}, \mathbb{Q}) \neq 0$, but $V^t(\boldsymbol{x}, \mathbb{Q}) = 0$. In Figure 3, we define a PPT adversary $\mathcal{B}_{\mathsf{fpr}}$ for the $(\boldsymbol{d}, m)$-FPR game. The assumption (6) ensures that the FPR oracle returns the desired answers. Since $V^t(\boldsymbol{X}, \mathbb{Q}) \neq 0$, there exists a nonzero component $g(\boldsymbol{X})$ as described above. Since $V^t(\boldsymbol{x}, \mathbb{Q}) = 0$, we have $g(\boldsymbol{x}) = 0$. By (6), $\deg(g(\boldsymbol{X})) \leq \max(md_1, md_2) \leq d_g$. Therefore, $\mathcal{B}_{\mathsf{fpr}}$ is successful. By the $(\boldsymbol{d}, m)$-FPR assumption, this can only happen with negligible probability.

Case Q**:** $V^t(\boldsymbol{x}, \mathbb{Q}) \neq 0$. In Figure 4, we define a PPT adversary $\mathcal{B}_{\mathsf{tofr}}$ for the $(\mathcal{EF}, \mathcal{DF})$-$\mathsf{TOFR}$ game. By Equation (8), $V^t(\boldsymbol{x}, \mathbb{Q}) \neq 0$ implies $\boldsymbol{v} \neq \boldsymbol{0}$ and $V(\boldsymbol{x}, \mathbb{q}) = V^h(\boldsymbol{x}) + V^t(\boldsymbol{x}, \mathbb{q}) = 0$ implies

$$\boldsymbol{v}^T \begin{pmatrix} 1 \\ \mathbb{q}_1 \\ \mathbb{q}_2 \\ \mathbb{q}_1 \otimes \mathbb{q}_2 \end{pmatrix} = 0.$$

Therefore, $\mathcal{B}_{\mathsf{tofr}}$ succeeds. By the $(\mathcal{EF}, \mathcal{DF})$-$\mathsf{TOFR}$ assumption, this can only happen with negligible probability. ☐

Remark 3. If we set $\mathsf{aux}_1(\boldsymbol{X})$ and $\mathsf{aux}_2(\boldsymbol{X})$ to empty strings, our lifting theorem (Theorem 1) *covers publicly verifiable assumptions*. Specifically, the $\mathsf{aux}_\kappa(\boldsymbol{X})$ terms simply vanish from the reduction, yet the argument remains identical.

3.3 Example: Extractability of the KZG polynomial commitment

To illustrate their AGMOS proof technique, Lipmaa et al. showed [LPS23, Section 5.3] that if the FPR and TOFR assumptions hold (for certain parameters), the univariate KZG polynomial commitment scheme (PCS) [KZG10] is extractable in the AGMOS. To demonstrate our framework for publicly verifiable games, we apply the notation from Section 3.1 to the extractability assumption for univariate KZG PCS.

Recall that, in a PCS, a prover commits to a polynomial $f(X)$ by computing a commitment $C = \mathsf{Com}(\mathsf{ck}, f(X))$, where ck is the commitment key. Later, the verifier issues a challenge evaluation point α, and the prover responds with the claimed evaluation $\eta = f(\alpha)$ and an associated opening proof π. The verifier then checks $\mathsf{Ver}(\mathsf{ck}, C, \alpha, \eta, \pi)$, accepting if and only if the proof π correctly demonstrates that the committed polynomial evaluated at α equals η.

We briefly recall how the KZG PCS works. The commitment key is given by $\mathsf{ck} = \left([x^i]_1\right)_{i=0}^{d}, \quad [1]_2,\ [x]_2$, which allows the prover to commit to univariate polynomials of degree at most d. To commit to a polynomial $f(X) = \sum_{i=0}^{d} f_i X^i$, the prover computes $C = \sum_{i=0}^{d} f_i[x^i]_1 = [f(x)]_1$, and sends the commitment C to the verifier. The verifier then selects a challenge point α and sends it to the prover. The prover responds by computing the claimed evaluation $\eta = f(\alpha)$ and the corresponding proof $\pi = [q(x)]_1$, where $q(X) = \frac{f(X)-\eta}{X-\alpha}$. If the prover is honest, the evaluation claim holds and they can compute the quotient polynomial $q(X)$, and, thus, provide a valid proof π. To verify the proof, the verifier checks the following pairing equation: $(C - [\eta]_1) \bullet [1]_2 = \pi \bullet [x - \alpha]_2$. If the equation holds, the verifier accepts the proof; otherwise, they reject.

Extractability for a PCS is a knowledge assumption: it requires that any adversary who convinces the verifier, i.e., produces an evaluation claim (C, α, η) with a proof π such that $\mathsf{Ver}(\mathsf{ck}, C, \alpha, \eta, \pi) = 1$, must "know" a polynomial $f(X)$ of degree at most d satisfying $\mathsf{Com}(\mathsf{ck}, f(X)) = C$ and $f(\alpha) = \eta$. Formally, this is captured by the following definition.

Definition 6 (Extractability of a PCS in the AGMOS). *A polynomial commitment scheme $\Pi = (\mathsf{KGen}, \mathsf{Com}, \mathsf{Open}, \mathsf{Vrf})$ is* extractable for PGen*, if for all $d \in \mathsf{poly}(\lambda)$ and every PPT adversary $\mathcal{A}$, there exists a PPT extractor $\mathsf{Ext}_{\mathcal{A}}$, such that*

$$\Pr\left[\begin{array}{l|l} \mathsf{Ver}(\mathsf{ck}, C, \alpha, \eta, \pi) = 1 \wedge & \mathsf{gp} \leftarrow \mathsf{PGen}(1^\lambda); \mathsf{ck} \leftarrow \mathsf{KPGen}(\mathsf{gp}, d); \\ (C \neq \mathsf{Com}(\mathsf{ck}, f(X)) \vee & ((C, \pi, \boldsymbol{\gamma}_1, \boldsymbol{\delta}_1, [\mathbb{q}_1]_1), \alpha, \eta) \leftarrow \mathcal{A}^{\mathcal{O}}(\mathsf{ck}); \\ \ \deg f(X) > d \ \vee\ f(\alpha) \neq \eta) & f(X) \leftarrow \mathsf{Ext}_{\mathcal{A}}(\mathsf{ck}, (C, \pi, \boldsymbol{\gamma}_1, \boldsymbol{\delta}_1, [\mathbb{q}_1]_1)) \end{array}\right] \in \mathsf{negl}(\lambda).$$

Note that, even though the extractor is provided the algebraic explanation of the commitment, the above definition is not vacuous in the AGM. In particular, one needs to prove that the polynomial output by the extractor matches the evaluation claim of the adversary. Some works in the literature consider also a stronger notion of *knowledge soundness of a PCS in the AGM(OS)*, where the extractor must construct the polynomial f *only* from the commitment and its algebraic explanation, but the rest of the security experiment is unchanged.

Next, we show that both extractability and knowledge soundness for the KZG polynomial commitment can be lifted via our lifting theorem from the AGM to the AGMOS. The proof follows by a straightforward verification of the lifting condition and a direct application of Theorem 1.

Theorem 2. *Assume that the $(\boldsymbol{d}, n)$-FPR and $(\mathcal{EF}, \mathcal{DF})$-$\mathsf{TOFR}$ assumptions hold, where $\boldsymbol{d} = (d, 1, d+1, d+1)$. If KZG PCS is extractable, respectively knowledge sound, in the AGM for univariate polynomials of degree at most d then it is extractable, respectively knowledge sound, in the $(\mathcal{EF}, \mathcal{DF})$-AGMOS for univariate polynomials of degree at most d.*

Proof. First, note that the restriction on the power of the extractor has no effect on the pairing verification condition. Thus, the verification of the lifting condition proceeds the same for extractability and knowledge soundness. Next, we make explicit the parameters of the security game.

For an AGMOS adversary, we have $m = 1$ ($\boldsymbol{x} = x_1 = x$), $\mathsf{il}_1 = d+1$, $\mathbb{x}_1(X) = (X^i)_{i=0}^{d}$, $\mathsf{il}_2 = 2$, $\mathbb{x}_2(X) = (1, X)$, $\mathsf{ol}_1 = 2$, $[\mathbb{y}_1]_1 = [\mathbb{y}_{11}, \mathbb{y}_{12}]_1 = (C, \pi)$, $\mathsf{ol}_2 = 0$, $r = 2$, $\boldsymbol{\alpha} = (\alpha, \eta)$. The explicit verification polynomial is obtained by rewriting the KZG pairing check $(C - [\eta]_1) \bullet [1]_2 = \pi \bullet [x - \alpha]_2$ in terms of discrete logarithms and rearranging all terms to one side. The resulting verification polynomial has the following form:

$$
\begin{aligned}
V^{\mathsf{expl}}(X, \mathbb{Y}_{11}, \mathbb{Y}_{12}) &= \mathbb{Y}_{11} - \eta - \mathbb{Y}_{12}(X - \alpha) \\
&= \begin{pmatrix} 1 & X & \dots & X^d & \mathbb{Y}_{11} & \mathbb{Y}_{12} \end{pmatrix} \underbrace{\begin{pmatrix} -\eta & 0 \\ \vdots & \vdots \\ 0 & 0 \\ 1 & 0 \\ \alpha & -1 \end{pmatrix}}_{\boldsymbol{M}} \begin{pmatrix} 1 \\ X \end{pmatrix}.
\end{aligned}
$$

The adversary $\mathcal{A}$ outputs a vector $[\mathbb{y}_1]_1 \in \mathbb{G}_1^2$ that admits a decomposition of the form $[\mathbb{y}_1]_1 = \boldsymbol{\gamma}_1[\mathbb{x}_1(\boldsymbol{x})]_1 + \boldsymbol{\delta}_1[\mathbb{q}_1]_1$, where $[\mathbb{q}_1]_1 = \left([\mathbb{q}_{11}]_1, \dots, [\mathbb{q}_{1\mathsf{ql}_1}]_1\right)$, and $\boldsymbol{\gamma}_1$ and $\boldsymbol{\delta}_1$ are matrices whose rows consist of the adversary's explanation vectors corresponding to the input elements $[\mathbb{x}_1(\boldsymbol{x})]_1$ and the obliviously sampled elements $[\mathbb{q}_1]_1$, respectively. Equivalently, in the discrete logarithm representation, the equation $[\mathbb{y}_1]_1 = \boldsymbol{\gamma}_1[\mathbb{x}_1(\boldsymbol{x})]_1 + \boldsymbol{\delta}_1[\mathbb{q}_1]_1$ can be rewritten as

$$
\mathbb{y}_1 = \boldsymbol{\gamma}_1 \mathbb{x}_1(x) + \boldsymbol{\delta}_1 \mathbb{q}_1 = \begin{pmatrix} \gamma_{1,1} & \dots & \gamma_{1,d+1} \\ \gamma_{2,1} & \dots & \gamma_{2,d+1} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ x^d \end{pmatrix} + \begin{pmatrix} \delta_{1,1} & \dots & \delta_{1,\mathsf{ql}_1} \\ \delta_{2,1} & \dots & \delta_{2,\mathsf{ql}_1} \end{pmatrix} \begin{pmatrix} \mathbb{q}_{11} \\ \vdots \\ \mathbb{q}_{1\mathsf{ql}_1} \end{pmatrix}.
$$

We now construct the implicit verification polynomial

$$
V(X, \mathbb{Q}) = \sum_{i=0}^{d} \gamma_{1,i+1} X^i + \sum_{i=1}^{\mathsf{ql}_1} \delta_{1i} \mathbb{Q}_{1i} - \eta - \left(\sum_{i=0}^{d} \gamma_{2,i+1} X^i + \sum_{i=1}^{\mathsf{ql}_1} \delta_{2,i} \mathbb{Q}_{1i} \right) (X - \alpha).
$$

By isolating the terms involving $\mathbb{Q}$ variables, we split the polynomial $V(X, \mathbb{Q})$ into two parts: one containing no $\mathbb{Q}$ variables, and the other consisting solely of terms involving $\mathbb{Q}$ variables. The polynomial $V^t(X, \mathbb{Q})$ has the following form:

$$
V^t(X, \mathbb{Q}) = \sum_{i=1}^{\mathsf{ql}_1} (\delta_{1,i} + \alpha \delta_{2,i}) \mathbb{Q}_{1i} - X \sum_{i=1}^{\mathsf{ql}_1} \delta_{2i} \mathbb{Q}_{1i}.
$$

Now, it is straightforward to verify the lifting condition from Definition 5, i.e. $V^t(X, \mathbb{Q}) \Rightarrow \boldsymbol{\delta} = \mathbf{0}$. Considering all the monomials $X\mathbb{Q}_{1i}$ and $\mathbb{Q}_{1i}$ appearing in $V^t(X, \mathbb{Q})$, we construct a system of linear equations

$$
\begin{aligned}
\delta_{1,i} + \alpha \delta_{2,i} &= 0 \qquad \forall 1 \leq i \leq \mathsf{ql}_1, \\
\delta_{2,i} &= 0 \qquad \forall 1 \leq i \leq \mathsf{ql}_1,
\end{aligned}
$$

which clearly admits only the trivial solution $\boldsymbol{\delta} = \mathbf{0}$. Therefore, we can invoke Theorem 1 to conclude that, under the assumed variants of FPR and TOFR, any AGM proof of extractability for the univariate KZG PCS implies the extractability also in the AGMOS. ◻

Lifting for other pairing-based PCS. Even though specific to the univariate KZG scheme, the above check of the lifting condition for univariate KZG can be readily extended to other pairing-based PCS due to the algebraic structure of standard pairing verification equations. In most pairing-based PCS, the verifier checks a polynomial identity by computing pairings between the adversary's output group elements (e.g., commitments and opening proofs) and group elements constructed from the commitment key based on the challenge values and prover's evaluation claims. In the discrete logarithm representation, this translates to an explicit verification polynomial V^{expl} that is strictly linear in the variables representing the adversary's outputs. Consequently, when substituting the AGMOS representation, the resulting $\mathbb{Q}$-dependent component, $V^t(X, \mathbb{Q})$, remains a strictly linear combination of the oblivious sampling coefficients $\delta_{j,i}$. Because these coefficients are multiplied by algebraically independent monomials in X and $\mathbb{Q}$ derived from the public parameters (analogous to the $X\mathbb{Q}_{1i}$ and $\mathbb{Q}_{1i}$ terms in the KZG proof), setting $V^t(X, \mathbb{Q}) = 0$ inherently yields a homogeneous system of linear equations. As long as the public polynomials involved in the pairing check are linearly independent, this system will only admit the trivial solution $\boldsymbol{\delta} = \mathbf{0}$. This satisfies the structural lifting condition, ensuring that the adversary's explanation relies solely on challenger-provided inputs. Therefore, any AGM security proof for such schemes can be lifted generically to the AGMOS under the FPR and TOFR assumptions.

3.4 Limitations of Our Lifting Theorem

The condition on $V^t(\boldsymbol{X}, \mathbb{Q})$ from Theorem 1 does not always hold. A trivial example is when $\boldsymbol{M} = \mathbf{0}$, i.e., no verification is performed. More generally, if $\boldsymbol{f_M}(\boldsymbol{\gamma}, \boldsymbol{\delta}) = \mathbf{0}$, then $V^t(\boldsymbol{X}, \mathbb{Q}) = 0$ (see Equation (3) and Equation (4)). The converse implication holds when $\mathrm{x}_\kappa(\boldsymbol{X})$ is linearly independent over $\mathbb{F}$ for $\kappa \in \{1, 2\}$. Lipmaa et al. analyzed the following family of Total Knowledge of Exponent (TotalKE) assumptions, which are trivially secure in the AGM.

Definition 7 (TotalKE). *Let* $\mathsf{il}_1 = \mathsf{il}_2 = 1, \mathsf{ol}_1, \mathsf{ol}_2 \geq 1$ *and* $R \geq 1$. *Let* $\boldsymbol{M}[i] \in \mathbb{F}^{(\mathsf{il}_1+\mathsf{ol}_1)\times(\mathsf{il}_2+\mathsf{ol}_2)}$ *for* $i \in [1, R]$. *Let*

$$V_i(\mathrm{y}) = \begin{pmatrix} 1 \\ \mathrm{y}_1 \end{pmatrix}^T \boldsymbol{M}[i] \begin{pmatrix} 1 \\ \mathrm{y}_2 \end{pmatrix}.$$

The $(\mathsf{ol}_1, \mathsf{ol}_2, \{V_i\}_{i=1}^R)$*-TotalKE assumption holds for* $\mathsf{PGen}(1^\lambda)$*, if for every non-uniform adversary* $\mathcal{A}$*, there exists a non-uniform PPT extractor* E*, such that*

$$\Pr\left[\begin{array}{l|l} \boldsymbol{y} \in \mathbb{F}^{\mathsf{ol}_1} \ \wedge \ \boldsymbol{z} \in \mathbb{F}^{\mathsf{ol}_2} \ \wedge & \mathsf{gp} \leftarrow \mathsf{PGen}(1^\lambda); r \leftarrow \mathsf{RND}_\lambda(\mathcal{A}); \\ \forall i \in [1, R]. V_i(\boldsymbol{y}, \boldsymbol{z}) = 0 \ \wedge & ([\boldsymbol{y}]_1, [\boldsymbol{z}]_2) \leftarrow \mathcal{A}(\mathsf{gp}, r); \\ (\boldsymbol{y}, \boldsymbol{z}) \neq (\boldsymbol{y}^*, \boldsymbol{z}^*) & ([\boldsymbol{y}^*]_1, [\boldsymbol{z}^*]_2) \leftarrow E_{\mathcal{A}}(\mathsf{gp}, r) \end{array}\right] \in \mathsf{negl}(\lambda).$$

They hypothesized that *"[...] the TotalKE assumption is secure [in the AGMOS under the* DLOG *assumption] iff for any* $\boldsymbol{\Gamma} = \boldsymbol{\gamma}$*,* $\boldsymbol{f_M}(\boldsymbol{\Gamma}, \boldsymbol{\Delta}) = \mathbf{0}$ *has only a zero solution in* $\boldsymbol{\Delta}$*."* Generalizing and proving a similar statement for our class of security games is an interesting line for future research.

3.5 Example: $\mathsf{ARSDH}(n)$ assumption

In this section, we illustrate our lifting theorem on a privately verifiable assumption. The Adaptive Rational Strong Diffie-Hellman assumption, ARSDH, was recently used in proofs of extractability of variants of the KZG PCS in the standard model [LPS24, BDH+25]. The univariate ARSDH assumption was proposed by Lipmaa et al. [Lip24], who established its security in the AGMOS. Subsequently, Belohorec et al. [BDH+25] introduced its multivariate variant $\mathsf{ARSDH}(n)$, and proved its security in the AGM. In this subsection, we use the results from Section 3.2 to lift the proof of security of the $\mathsf{ARSDH}(n)$ assumption in the AGM [BDH+25] to a proof of security in the AGMOS.

We start with defining the assumption.

Definition 8 ($\mathsf{ARSDH}(n)$ **Game).** *For a PPT adversary* $\mathcal{A}$*, a bilinear group generator* PGen*, and* $\mathbf{d}' = (d'_1, \ldots, d'_n) \in \mathsf{poly}(\lambda)$*, the* $\mathsf{ARSDH}(n)$ Game *is defined as follows:*

1. *Sample* $\mathsf{gp} \leftarrow \mathsf{PGen}(1^\lambda)$ *and* $\boldsymbol{x} = (x_1, \ldots, x_n) \leftarrow \mathbb{F}^n$.
2. *On input* $1^\lambda, \mathsf{gp}$*, and* $\left(\left([x_1^{i_1} \cdots x_n^{i_n}]_1\right)_{i_1,\ldots,i_n=0}^{d'_1,\ldots,d'_n}, ([x_i]_2)_{i\in[1,n]}\right)$, $\mathcal{A}$ *outputs*

$$\{[\mathrm{y}_i]_1\}_{i=0}^n, \quad \{\alpha_i\}_{i=1}^{n-1} \subseteq \mathbb{F}, \quad \textit{and} \quad S \subseteq \mathbb{F}.$$

$\mathcal{A}$ *wins if and only if:*

1. $|S| = d'_n + 1$,
2. $[\mathbb{y}_0]_1 \bullet [1]_2 \neq \bigodot_{i=1}^{n-1} [\mathbb{y}_i]_1 \bullet [x_i - \alpha_i]_2$, *and*
3. $[\mathbb{y}_0]_1 \bullet [1]_2 = \left(\bigodot_{i=1}^{n-1} [\mathbb{y}_i]_1 \bullet [x_i - \alpha_i]_2\right) \odot [\mathbb{y}_n]_1 \bullet [Z_S(x_n)]_2$.

We write $\mathsf{ARSDH}(n).\mathsf{Game}(\mathcal{A}, \mathsf{PGen}, \mathbf{d}') = 1$ *if* $\mathcal{A}$ *wins and 0 otherwise.*

The polynomial $Z_S(X)$ is the vanishing polynomial of the set S, defined as $Z_S(X) = \prod_{s \in S}(X - s)$.

Definition 9 **($\mathsf{ARSDH}(n)$).** *Let* PGen *be a bilinear group generator. Then the* n*-variate* Adaptive Rational Strong Diffie-Hellman ($\mathsf{ARSDH}(n)$) assumption *holds for* PGen *if, for any* $\mathbf{d}' = (d'_1, \ldots, d'_n) \in \mathsf{poly}(\lambda)$ *and PPT* $\mathcal{A}$,

$$\Pr[\mathsf{ARSDH}(n).\mathsf{Game}(\mathcal{A}, \mathsf{PGen}, \mathbf{d}') = 1] \in \mathsf{negl}(\lambda).$$

Next, we lift the security of the $\mathsf{ARSDH}(n)$ from AGM to AGMOS using Theorem 1.

Theorem 3 **(Lifting $\mathsf{ARSDH}(n)$ to AGMOS).** *Assume that* $(\boldsymbol{d}, n)$-FPR *and* $(\mathcal{EF}, \mathcal{DF})$-$\mathsf{TOFR}$ *assumptions hold, where* $\boldsymbol{d} = (\max_i\{d'_i\}, 1, \max_i\{d'_i\} + 1, \sum_{i=1}^{n} d'_i + d'_n + 1)$. *Also, assume that the* $\mathsf{ARSDH}(n)$ *assumption is secure in the AGM. Then the* $\mathsf{ARSDH}(n)$ *assumption is secure in the* $(\mathcal{EF}, \mathcal{DF})$*-AGMOS.*

Proof. In the notation of Theorem 1, we have $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_{n-1}, S)$, and, hence $r = d'_n + n$, $\mathsf{il}_1 = (d'_1 + 1) \cdots (d'_n + 1), \mathsf{il}_2 = n + 1$, $\mathsf{ol}_1 = n + 1, \mathsf{ol}_2 = 0$. In this case, $\mathsf{aux}_1(\boldsymbol{X})$ is an empty string and $\mathsf{aux}_2(\boldsymbol{X}) = (X_n^2, \ldots, X_n^{d'_n+1}) \in \mathbb{F}[\boldsymbol{X}]^{d'_n}$.

The explicit verification polynomial is of the following form:

$$V^{\mathsf{expl}}(X_1, \ldots, X_n, \mathbb{Y}_{10}, \ldots, \mathbb{Y}_{1n}) = \mathbb{Y}_{10} - \sum_{i=1}^{n-1} \mathbb{Y}_{1i} \cdot (X_i - \alpha_i) - \mathbb{Y}_{1n} \cdot Z_S(X_n)$$

$$= \left(\mathbb{x}_1^T(\boldsymbol{X})\ \mathbb{Y}_{10}\ \mathbb{Y}_{11}\ \mathbb{Y}_{12} \cdots \mathbb{Y}_{1n-1}\ \mathbb{Y}_{1n}\right) \underbrace{\begin{pmatrix} \multicolumn{9}{c}{\mathbf{0}_{\mathsf{il}_1 \times (d'_n + n + 1)}} \\ \hline 1 & 0 & 0 & \ldots & 0 & 0 & 0 & \ldots & 0 \\ \alpha_1 & -1 & 0 & \ldots & 0 & 0 & 0 & \ldots & 0 \\ \alpha_2 & 0 & -1 & \ldots & 0 & 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & & \vdots \\ \alpha_{n-1} & 0 & 0 & \ldots & -1 & 0 & 0 & \ldots & 0 \\ -z_0 & 0 & 0 & \ldots & 0 & -z_1 & -z_2 & \ldots & -z_{d'_n+1} \end{pmatrix}}_{\boldsymbol{M}} \begin{pmatrix} 1 \\ X_1 \\ X_2 \\ \vdots \\ X_{n-1} \\ X_n \\ X_n^2 \\ \vdots \\ X_n^{d'_n+1} \end{pmatrix},$$

where the polynomial $Z_S(X_n) = \sum_{j=0}^{d'_n+1} z_j X_n^j$ and $\mathbb{Y}_{1i}$ for $i \in [0, n]$ are variables that correspond to discrete logarithms of group elements output by the adversary $\mathcal{A}$. The adversary $\mathcal{A}$ produces a vector $[\mathbb{y}_1]_1 \in \mathbb{G}_1^{n+1}$ which admits a decomposition of the form $\boldsymbol{\gamma}_1[\mathbb{x}_1(\boldsymbol{x})]_1 + \boldsymbol{\delta}_1[\mathbb{q}_1]_1$, where $[\mathbb{q}_1]_1 = [\mathbb{q}_{11}, \ldots, \mathbb{q}_{1\mathsf{ql}_1}]_1^T$ and $\boldsymbol{\gamma}_1, \boldsymbol{\delta}_1$ are matrices formed by the explanation vectors. Equivalently, in the discrete logarithm representation, the equation $[\mathbb{y}_1]_1 = \boldsymbol{\gamma}_1[\mathbb{x}_1(\boldsymbol{x})]_1 + \boldsymbol{\delta}_1[\mathbb{q}_1]_1$ looks like this:

$$\mathbb{y}_1 = \boldsymbol{\gamma}_1 \mathbb{x}_1(\boldsymbol{x}) + \boldsymbol{\delta}_1 \mathbb{q}_1 = \begin{pmatrix} \gamma_{0,1} & \cdots & \gamma_{0,\mathsf{il}_1} \\ \vdots & & \vdots \\ \gamma_{n,1} & \cdots & \gamma_{n,\mathsf{il}_1} \end{pmatrix} \mathbb{x}_1(\boldsymbol{x}) + \begin{pmatrix} \delta_{0,1} & \cdots & \delta_{0,\mathsf{ql}_1} \\ \vdots & & \vdots \\ \delta_{n,1} & \cdots & \delta_{n,\mathsf{ql}_1} \end{pmatrix} \begin{pmatrix} \mathbb{q}_{11} \\ \vdots \\ \mathbb{q}_{1\mathsf{ql}_1} \end{pmatrix}.$$

We now construct the implicit verification polynomial that looks like this:

$$V(\boldsymbol{X}, \mathbb{Q}) = \mathbb{Y}_{10}(\boldsymbol{X}, \mathbb{Q}) - \sum_{i=1}^{n-1} \mathbb{Y}_{1i}(\boldsymbol{X}, \mathbb{Q}) \cdot (X_i - \alpha_i) - \mathbb{Y}_{1n}(\boldsymbol{X}, \mathbb{Q}) \cdot Z_S(X_n),$$

where $\mathbb{Y}_{1i}$ is the variable representing the discrete logarithm of the i-th coordinate of the output vector $[\mathbb{y}_1]_1$ and it can be written as a linear combination of $\mathbb{x}_1(\boldsymbol{X})$ and $\mathbb{Q}$, in the following way:

$$\mathbb{Y}_{1i}(\boldsymbol{X}, \mathbb{Q}) = \sum_{j=1}^{\mathsf{il}_1} \gamma_{i,j} \mathbb{x}_{1j}(\boldsymbol{X}) + \sum_{j=1}^{\mathsf{ql}_1} \delta_{i,j} \mathbb{Q}_{1j}, \ i \in [0, n],$$

where $\mathbb{x}_{1j}(\boldsymbol{X})$ is the j-th coordinate of the vector $\mathbb{x}_1(\boldsymbol{X})$.

When we rewrite the implicit verification polynomial in the terms of $\mathbb{x}_1(\boldsymbol{X})$ and $\mathbb{Q}_1$, we get:

$$\begin{aligned} V(\boldsymbol{X}, \mathbb{Q}) = & \sum_{j=1}^{\mathsf{il}_1} \gamma_{0,j} \mathbb{x}_{1j}(\boldsymbol{X}) + \sum_{j=1}^{\mathsf{ql}_1} \delta_{0,j} \mathbb{Q}_{1j} - \sum_{i=1}^{n-1} \left(\sum_{j=1}^{\mathsf{il}_1} \gamma_{i,j} \mathbb{x}_{1j}(\boldsymbol{X}) + \sum_{j=1}^{\mathsf{ql}_1} \delta_{i,j} \mathbb{Q}_{1j} \right) (X_i - \alpha_i) \\ & - \left(\sum_{j=1}^{\mathsf{il}_1} \gamma_{n,j} \mathbb{x}_{1j}(\boldsymbol{X}) + \sum_{j=1}^{\mathsf{ql}_1} \delta_{n,j} \mathbb{Q}_{1j} \right) Z_S(X_n). \end{aligned}$$

By isolating the terms involving $\mathbb{Q}$ variables, we split the polynomial $V(X, \mathbb{Q})$ into two parts: one containing no $\mathbb{Q}$ variables, and the other consisting solely of terms involving $\mathbb{Q}$ variables. The polynomial $V^t(\boldsymbol{X}, \mathbb{Q})$ has the following form:

$$\begin{aligned} V^t(\boldsymbol{X}, \mathbb{Q}) &= \sum_{j=1}^{\mathsf{ql}_1} \delta_{0,j} \mathbb{Q}_{1j} - \sum_{i=1}^{n-1} (X_i - \alpha_i) \left(\sum_{j=1}^{\mathsf{ql}_1} \delta_{i,j} \mathbb{Q}_{1j} \right) - \left(\sum_{j=1}^{\mathsf{ql}_1} \delta_{n,j} \mathbb{Q}_{1j} \right) Z_S(X_n) \\ &= \sum_{j=1}^{\mathsf{ql}_1} \mathbb{Q}_{1j} \left(\delta_{0,j} - \sum_{i=1}^{n-1} (X_i - \alpha_i) \delta_{i,j} - \delta_{n,j} Z_S(X_n) \right) \end{aligned} \tag{9}$$

Now, it is straightforward to verify the lifting condition. The coefficients of $\mathbb{Q}_{1j}$ in Equation (9) are $\delta_{0,j} - \sum_{i=1}^{n-1} (X_i - \alpha_i) \delta_{i,j} - \delta_{n,j} Z_S(X_n)$. Thus, $V^t(\boldsymbol{X}, \mathbb{Q}) = 0$ is equivalent to the following system of ql_1 linear equations in the variables $\delta_{i,j}$, where $i \in [0, n]$ and $j \in [1, \mathsf{ql}_1]$:

$$\delta_{0,j} - \sum_{i=1}^{n-1} (X_i - \alpha_i) \delta_{i,j} - \delta_{n,j} Z_S(X_n) = 0, \quad \forall 1 \leq j \leq \mathsf{ql}_1.$$

For each j, we can proceed from $\delta_{n,j}$ to $\delta_{0,j}$ and verify that it has to be zero for the corresponding monomial in X_i to vanish and use that to simplify the equation. Eventually, we conclude that $\delta_{i,j} = 0$ for all $i \in [0, n]$ and $j \in [1, \mathsf{ql}_1]$. Therefore, the lifting condition holds and we can invoke Theorem 1 to conclude that, under the specific FPR and TOFR assumptions, any proof of security for $\mathsf{ARSDH}(n)$ lifts from the AGM to the AGMOS. ◻

Acknowledgments

Juraj Belohorec and Pavel Hubáček were partially supported by the Academy of Sciences of the Czech Republic (RVO 67985840), Czech Science Foundation GAČR grant No. 25-16311S, and by Zircuit. Juraj Belohorec was supported by the grant SVV–2025–260822.

References

BDF+11. Dan Boneh, Özgür Dagdelen, Marc Fischlin, Anja Lehmann, Christian Schaffner, and Mark Zhandry. Random oracles in a quantum world. In Dong Hoon Lee and Xiaoyun Wang, editors, *ASIACRYPT 2011*, volume 7073 of *LNCS*, pages 41–69. Springer, Berlin, Heidelberg, December 2011.

BDFG21. Dan Boneh, Justin Drake, Ben Fisch, and Ariel Gabizon. Halo infinite: Proof-carrying data from additive polynomial commitments. In Tal Malkin and Chris Peikert, editors, *CRYPTO 2021, Part I*, volume 12825 of *LNCS*, pages 649–680, Virtual Event, August 2021. Springer, Cham.

BDH+25. Juraj Belohorec, Pavel Dvořák, Charlotte Hoffmann, Pavel Hubáček, Kristýna Mašková, and Martin Pastyřík. On extractability of the KZG family of polynomial commitment schemes. In Yael Tauman Kalai and Seny F. Kamara, editors, *Advances in Cryptology - CRYPTO 2025 - 45th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 17-21, 2025, Proceedings, Part VI*, volume 16005 of *Lecture Notes in Computer Science*, pages 584–616. Springer, 2025.

BF01. Dan Boneh and Matthew K. Franklin. Identity-based encryption from the Weil pairing. In Joe Kilian, editor, *CRYPTO 2001*, volume 2139 of *LNCS*, pages 213–229. Springer, Berlin, Heidelberg, August 2001.

BFHK24. Balthazar Bauer, Pooya Farshim, Patrick Harasser, and Markulf Kohlweiss. The uber-knowledge assumption: A bridge to the AGM. *IACR Commun. Cryptol.*, 1(3):31, 2024.

BR93. Mihir Bellare and Phillip Rogaway. Random oracles are practical: A paradigm for designing efficient protocols. In Dorothy E. Denning, Raymond Pyle, Ravi Ganesan, Ravi S. Sandhu, and Victoria Ashby, editors, *ACM CCS 93*, pages 62–73. ACM Press, November 1993.

BV98. Dan Boneh and Ramarathnam Venkatesan. Breaking RSA may not be equivalent to factoring. In Kaisa Nyberg, editor, *EUROCRYPT'98*, volume 1403 of *LNCS*, pages 59–71. Springer, Berlin, Heidelberg, May / June 1998.

CFF+24. Matteo Campanelli, Antonio Faonio, Dario Fiore, Tianyu Li, and Helger Lipmaa. Lookup arguments: Improvements, extensions and applications to zero-knowledge decision trees. In *Public-Key Cryptography - PKC 2024 - 27th IACR International Conference on Practice and Theory of Public-Key Cryptography, Sydney, NSW, Australia, April 15-17, 2024, Proceedings, Part II*, pages 337–369, 2024.

CHM+20. Alessandro Chiesa, Yuncong Hu, Mary Maller, Pratyush Mishra, Psi Vesely, and Nicholas P. Ward. Marlin: Preprocessing zkSNARKs with universal and updatable SRS. In Anne Canteaut and Yuval Ishai, editors, *EUROCRYPT 2020, Part I*, volume 12105 of *LNCS*, pages 738–768. Springer, Cham, May 2020.

DMS24. Michel Dellepere, Pratyush Mishra, and Alireza Shirzad. Garuda and pari: Faster and smaller SNARKs via equifficient polynomial commitments. *IACR Cryptol. ePrint Arch.*, page 1245, 2024.

FFR24. Antonio Faonio, Dario Fiore, and Luigi Russo. Real-world universal zkSNARKs are non-malleable. In Bo Luo, Xiaojing Liao, Jun Xu, Engin Kirda, and David Lie, editors, *ACM CCS 2024*, pages 3138–3151. ACM Press, October 2024.

Fis00. Marc Fischlin. A note on security proofs in the generic model. In Tatsuaki Okamoto, editor, *ASIACRYPT 2000*, volume 1976 of *LNCS*, pages 458–469. Springer, Berlin, Heidelberg, December 2000.

FKL18. Georg Fuchsbauer, Eike Kiltz, and Julian Loss. The algebraic group model and its applications. In Hovav Shacham and Alexandra Boldyreva, editors, *CRYPTO 2018, Part II*, volume 10992 of *LNCS*, pages 33–62. Springer, Cham, August 2018.

FPS20. Georg Fuchsbauer, Antoine Plouviez, and Yannick Seurin. Blind Schnorr signatures and signed ElGamal encryption in the algebraic group model. In Anne Canteaut and Yuval Ishai, editors, *EUROCRYPT 2020, Part II*, volume 12106 of *LNCS*, pages 63–95. Springer, Cham, May 2020.

GWC19. Ariel Gabizon, Zachary J. Williamson, and Oana Ciobotaru. PLONK: Permutations over Lagrange-bases for oecumenical noninteractive arguments of knowledge. Cryptology ePrint Archive, Report 2019/953, 2019.

Ica09. Thomas Icart. How to hash into elliptic curves. In Shai Halevi, editor, *CRYPTO 2009*, volume 5677 of *LNCS*, pages 303–316. Springer, Berlin, Heidelberg, August 2009.

JM24. Joseph Jaeger and Deep Inder Mohan. Generic and algebraic computation models: When AGM proofs transfer to the GGM. In *Advances in Cryptology - CRYPTO 2024 - 44th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 18-22, 2024, Proceedings, Part V*, pages 14–45, 2024.

JR10. Tibor Jager and Andy Rupp. The semi-generic group model and applications to pairing-based cryptography. In Masayuki Abe, editor, *Advances in Cryptology – ASIACRYPT 2010 – 16th International Conference on the Theory and Application of Cryptology and Information Security, Proceedings*, volume 6477 of *Lecture Notes in Computer Science*, pages 539–556. Springer, 2010.

KZG10. Aniket Kate, Gregory M. Zaverucha, and Ian Goldberg. Constant-size commitments to polynomials and their applications. In Masayuki Abe, editor, *ASIACRYPT 2010*, volume 6477 of *LNCS*, pages 177–194. Springer, Berlin, Heidelberg, December 2010.

Lip12. Helger Lipmaa. Progression-free sets and sublinear pairing-based non-interactive zero-knowledge arguments. In Ronald Cramer, editor, *TCC 2012*, volume 7194 of *LNCS*, pages 169–189. Springer, Berlin, Heidelberg, March 2012.

Lip24. Helger Lipmaa. Polymath: Groth16 is not the limit. In *Advances in Cryptology - CRYPTO 2024 - 44th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 18-22, 2024, Proceedings, Part X*, pages 170–206, 2024.

LPS23. Helger Lipmaa, Roberto Parisella, and Janno Siim. Algebraic group model with oblivious sampling. In Guy N. Rothblum and Hoeteck Wee, editors, *TCC 2023, Part IV*, volume 14372 of *LNCS*, pages 363–392. Springer, Cham, November / December 2023.

LPS24. Helger Lipmaa, Roberto Parisella, and Janno Siim. Constant-size zk-SNARKs in ROM from falsifiable assumptions. In Marc Joye and Gregor Leander, editors, *EUROCRYPT 2024, Part VI*, volume 14656 of *LNCS*, pages 34–64. Springer, Cham, May 2024.

LPS25. Helger Lipmaa, Roberto Parisella, and Janno Siim. On knowledge-soundness of Plonk in ROM from falsifiable assumptions. In Yael Tauman Kalai and Seny F. Kamara, editors, *Advances in Cryptology - CRYPTO 2025 - 45th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 17-21, 2025, Proceedings, Part VII*, volume 16006 of *Lecture Notes in Computer Science*, pages 362–395. Springer, 2025.

Mau05. Ueli M. Maurer. Abstract models of computation in cryptography. In Nigel P. Smart, editor, *Cryptography and Coding, 10th IMA International Conference, Cirencester, UK, December 19-21, 2005, Proceedings*, volume 3796 of *Lecture Notes in Computer Science*, pages 1–12. Springer, 2005.

MBKM19. Mary Maller, Sean Bowe, Markulf Kohlweiss, and Sarah Meiklejohn. Sonic: Zero-knowledge SNARKs from linear-size universal and updatable structured reference strings. In Lorenzo Cavallaro, Johannes Kinder, XiaoFeng Wang, and Jonathan Katz, editors, *ACM CCS 2019*, pages 2111–2128. ACM Press, November 2019.

NPR26. Anca Nitulescu, Nikitas Paslis, and Carla Ràfols. Flip-and-prove R1CS. *IACR Commun. Cryptol.*, 2(4):6, 2026.

PV05. Pascal Paillier and Damien Vergnaud. Discrete-log-based signatures may not be equivalent to discrete log. In Bimal K. Roy, editor, *ASIACRYPT 2005*, volume 3788 of *LNCS*, pages 1–20. Springer, Berlin, Heidelberg, December 2005.

Sho97. Victor Shoup. Lower bounds for discrete logarithms and related problems. In *Advances in Cryptology - EUROCRYPT '97, International Conference on the Theory and Application of Cryptographic Techniques, Konstanz, Germany, May 11-15, 1997, Proceeding*, pages 256–266, 1997.

WB19. Riad S. Wahby and Dan Boneh. Fast and simple constant-time hashing to the BLS12-381 elliptic curve. *IACR TCHES*, 2019(4):154–179, 2019.

WZ24. Benedikt Wagner and Arantxa Zapico. A documentation of ethereum's peerdas. *IACR Cryptol. ePrint Arch.*, page 1362, 2024.

ZZK22. Cong Zhang, Hong-Sheng Zhou, and Jonathan Katz. An analysis of the algebraic group model. In Shweta Agrawal and Dongdai Lin, editors, *ASIACRYPT 2022, Part IV*, volume 13794 of *LNCS*, pages 310–322. Springer, Cham, December 2022.